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Quader et al.

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(54) **DEVICE WITH EMBEDDED
HIGH-BANDWIDTH, HIGH-CAPACITY
MEMORY USING WAFER BONDING**

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(Continued)

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CPC **H01L 25/0657** (2013.01); **G06F 3/0604** (2013.01); **G06F 3/0655** (2013.01);
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CPC H01L 25/0657; H01L 25/50; H01L 2225/06513; H01L 2225/06541;
(Continued)

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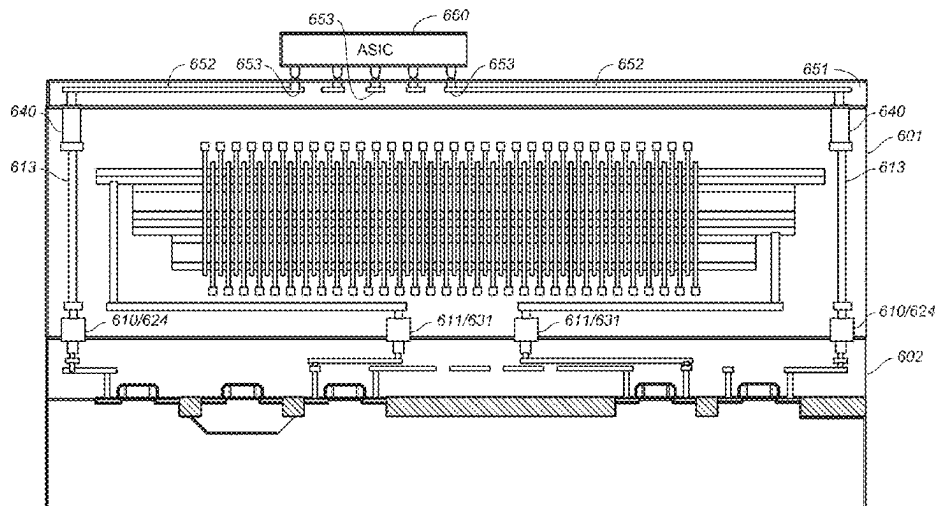
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Primary Examiner — Mamadou L Diallo

(57) **ABSTRACT**

An electronic device with embedded access to a high-bandwidth, high-capacity fast-access memory includes (a) a memory circuit fabricated on a first semiconductor die, wherein the memory circuit includes numerous modular memory units, each modular memory unit having (i) a three-dimensional array of storage transistors, and (ii) a group of conductors exposed to a surface of the first semiconductor die, the group of conductors being configured for communicating control, address and data signals associated the memory unit; and (b) a logic circuit fabricated on a second semiconductor die, wherein the logic circuit also includes conductors each exposed at a surface of the second semiconductor die, wherein the first and second semiconductor dies are wafer-bonded, such that the conductors exposed at the surface of the first semiconductor die are each electrically connected to a corresponding one of the conductors exposed to the surface of the second semiconductor die. The three-dimensional array of storage transistors may be formed by NOR memory strings.

67 Claims, 25 Drawing Sheets



Related U.S. Application Data

division of application No. 16/776,279, filed on Jan. 29, 2020, now Pat. No. 11,670,620.

- (60) Provisional application No. 62/843,733, filed on May 6, 2019, provisional application No. 62/803,689, filed on Feb. 11, 2019, provisional application No. 62/798,673, filed on Jan. 30, 2019.

(51) **Int. Cl.**

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G06N 3/02 (2006.01)
G11C 16/04 (2006.01)
H01L 25/00 (2006.01)

(52) **U.S. Cl.**

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2212/72; G06N 3/02; G11C 16/0483;
 G11C 16/3418

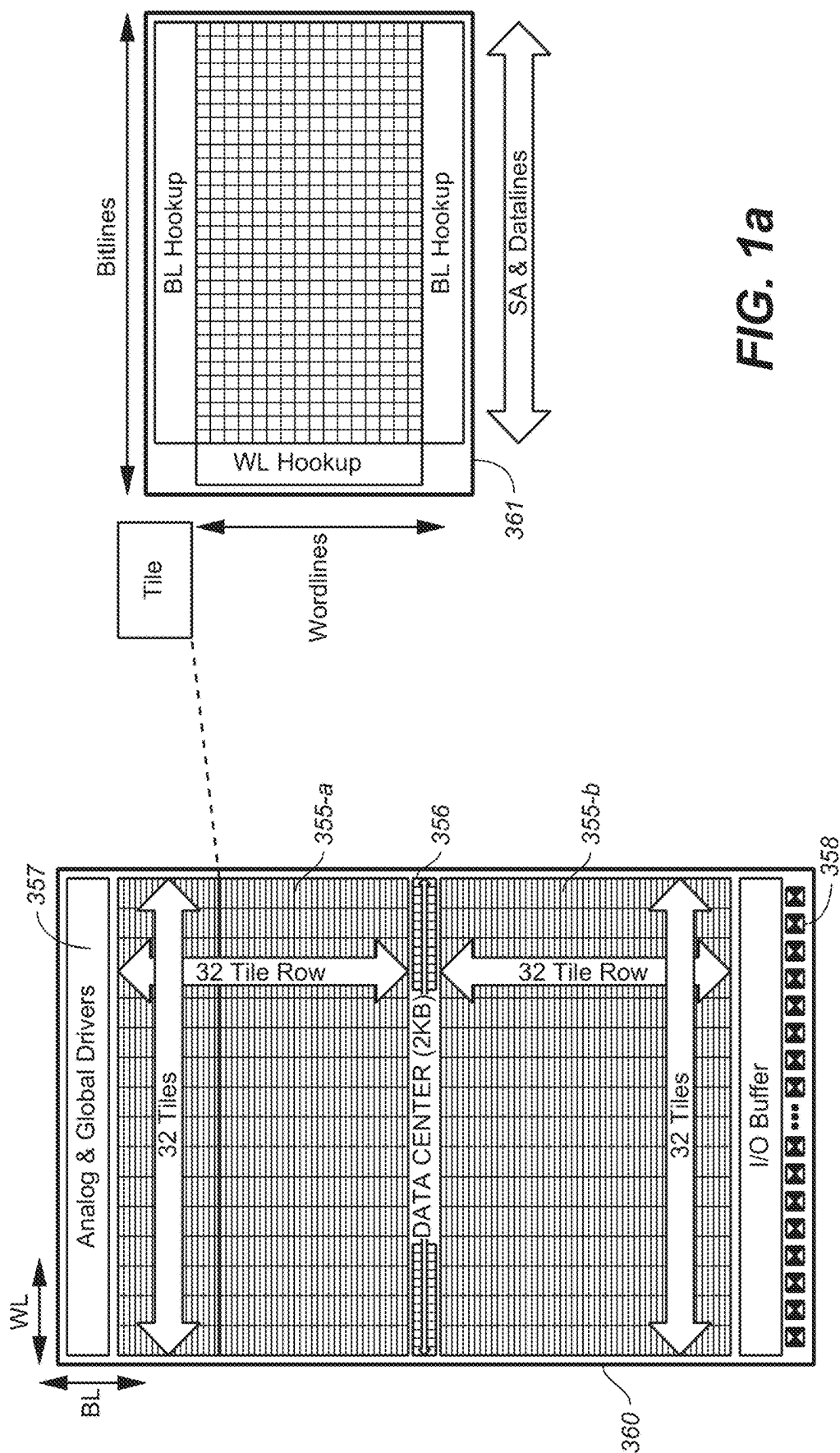
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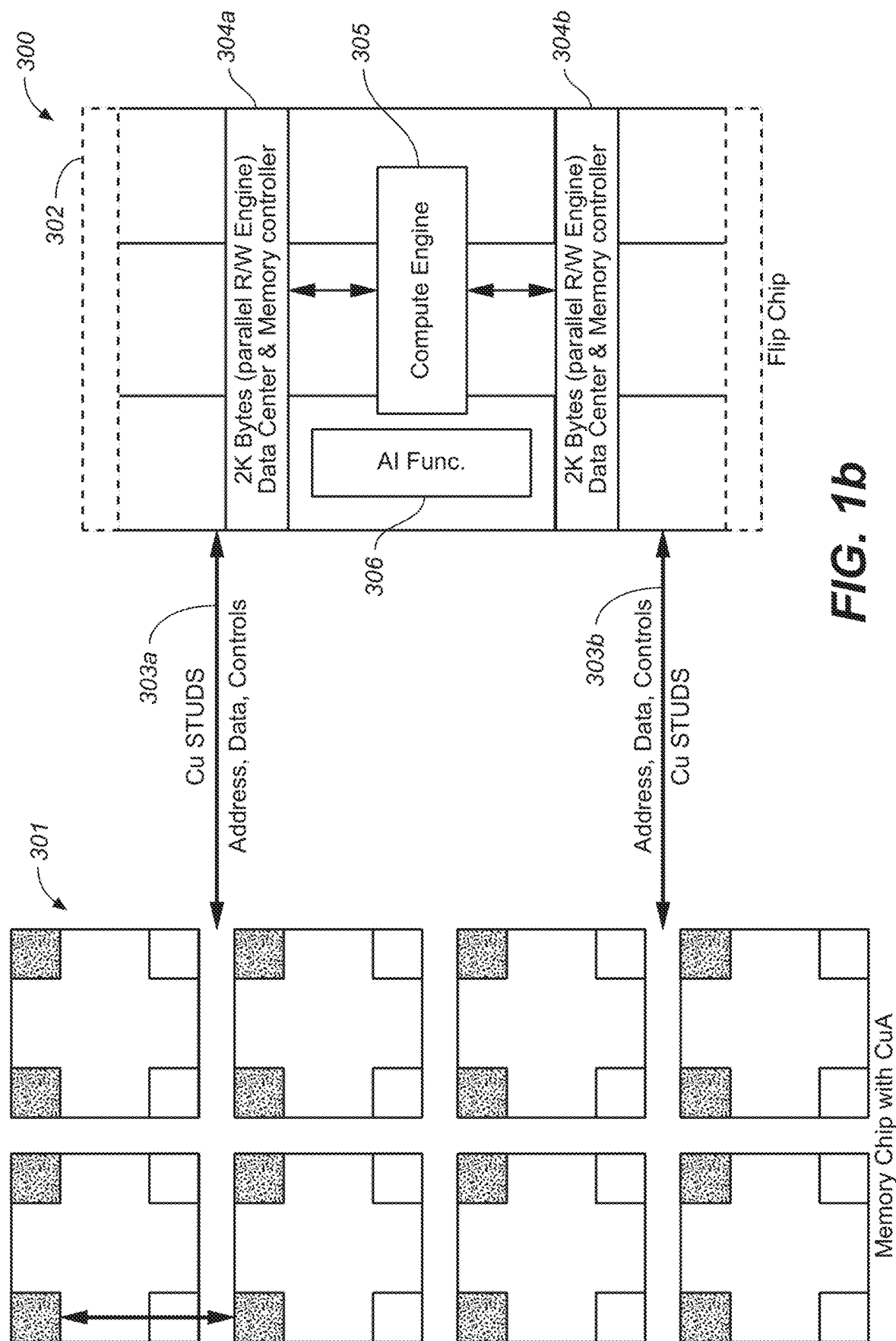
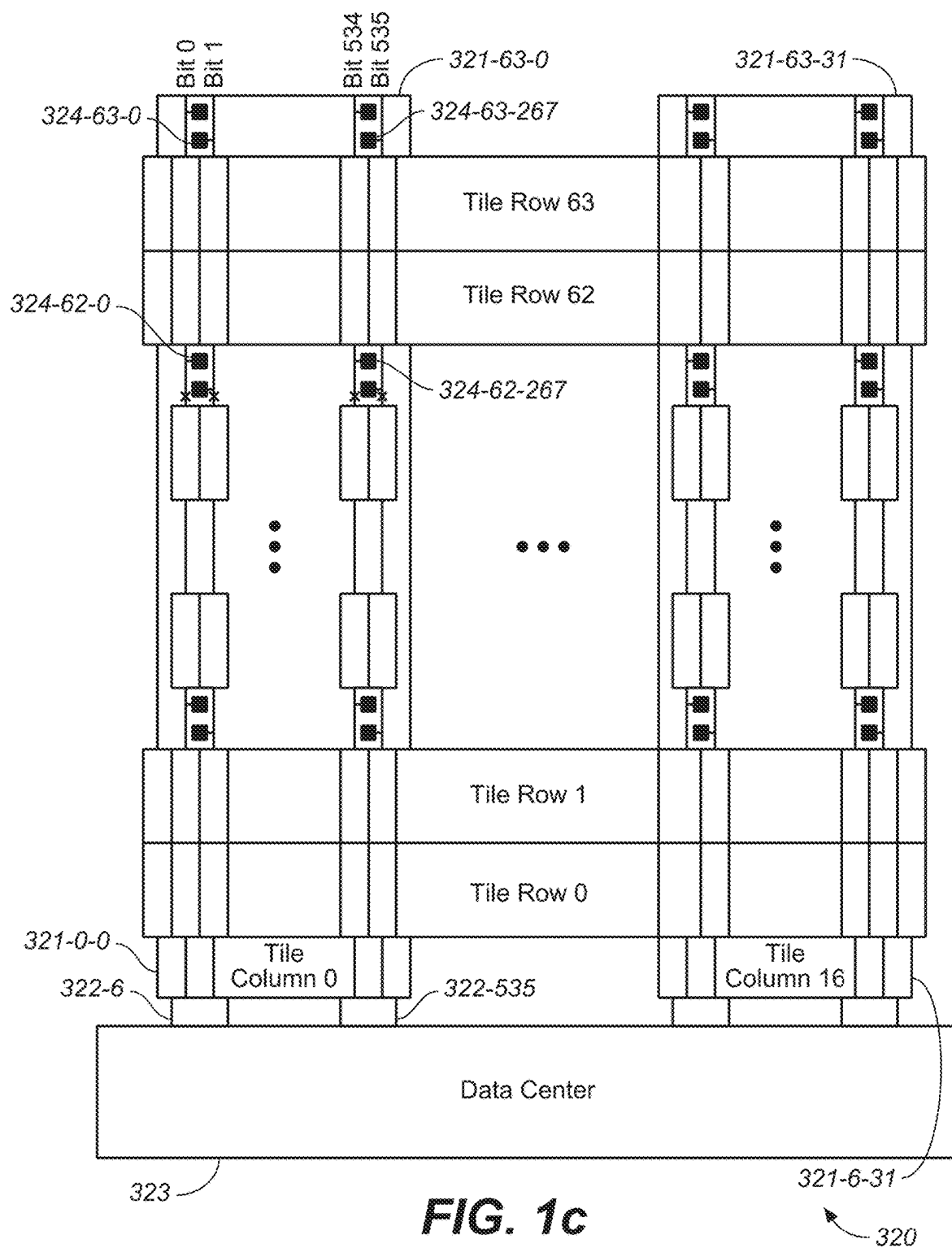


FIG. 1b



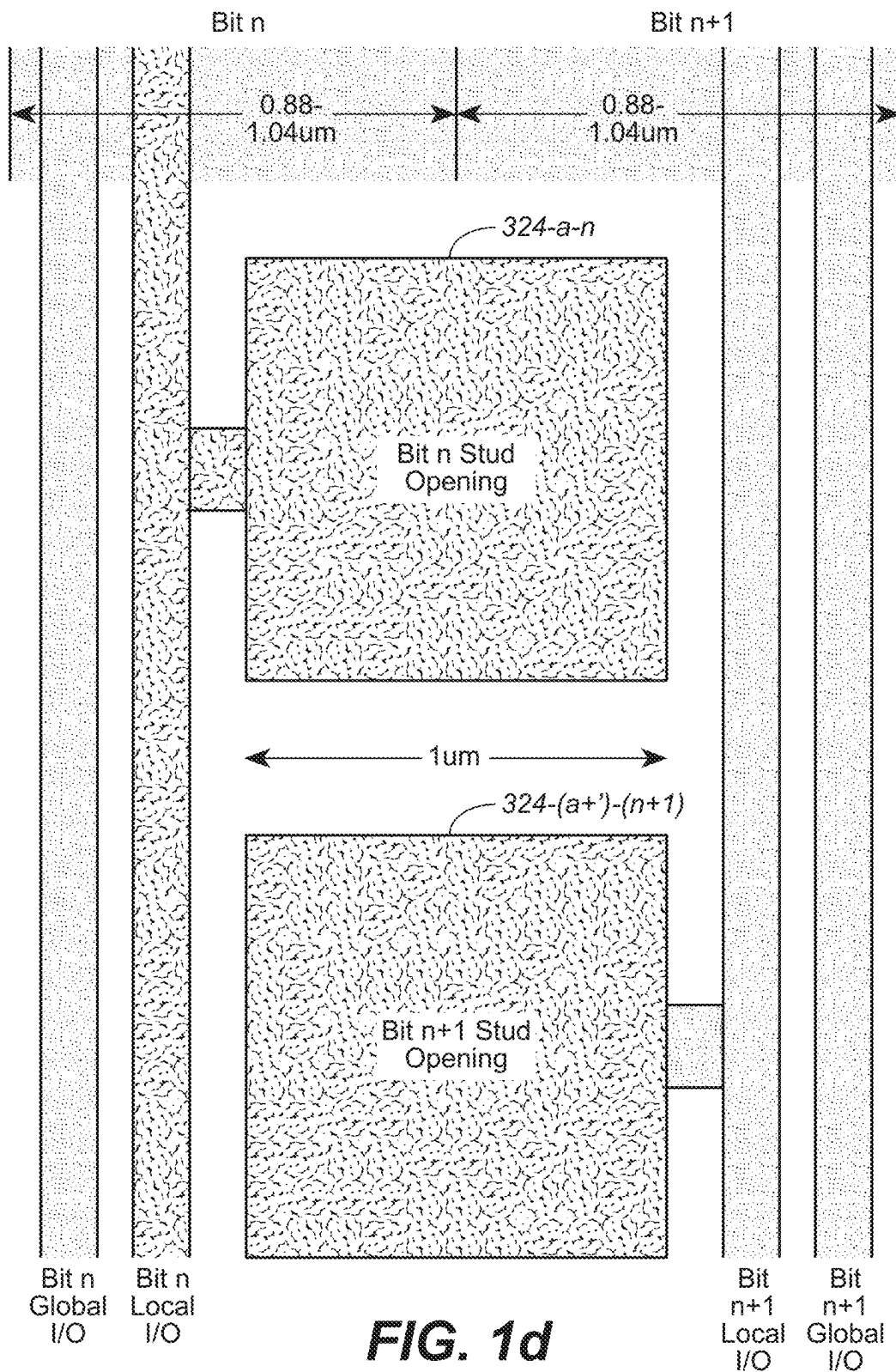
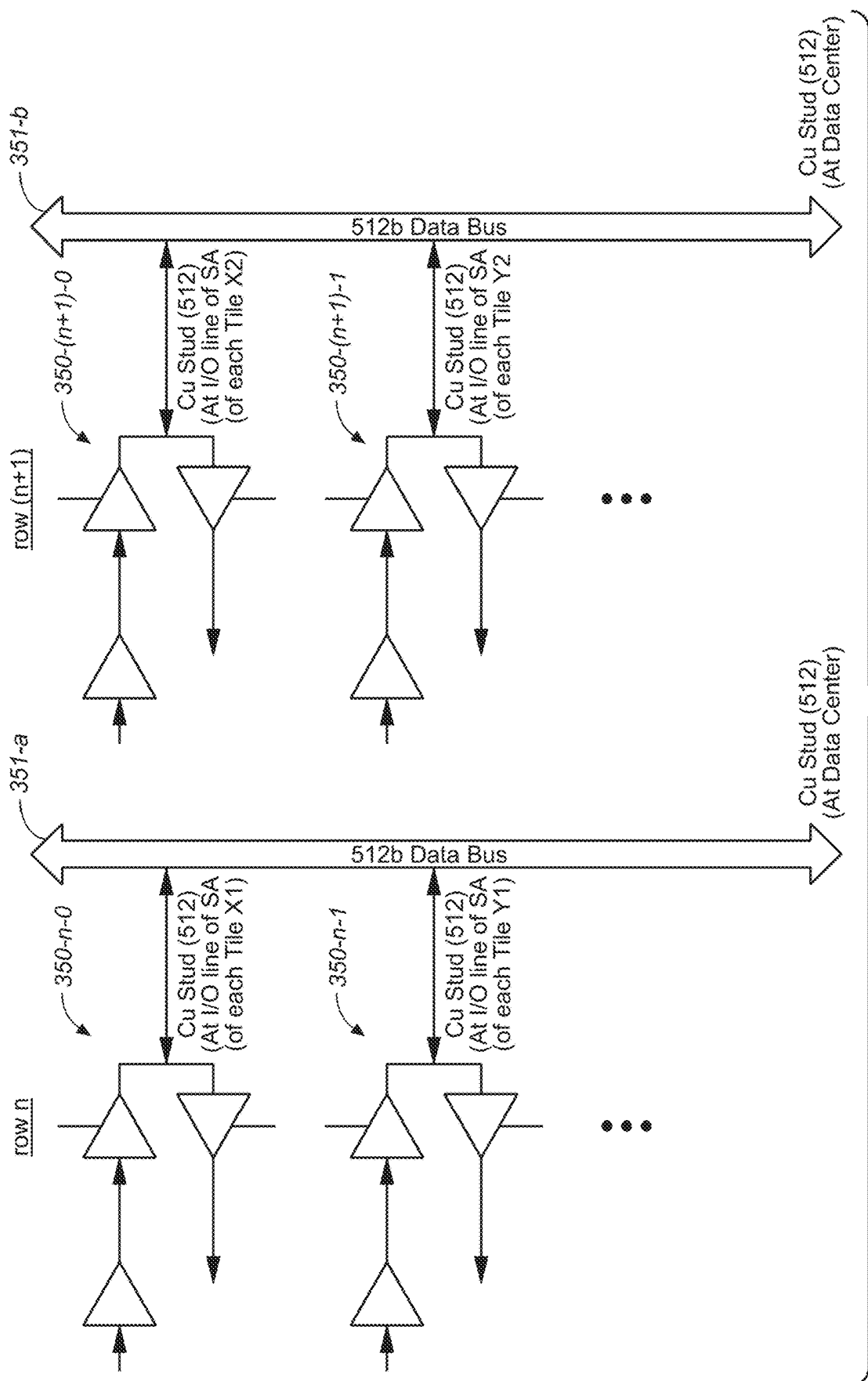


FIG. 1d



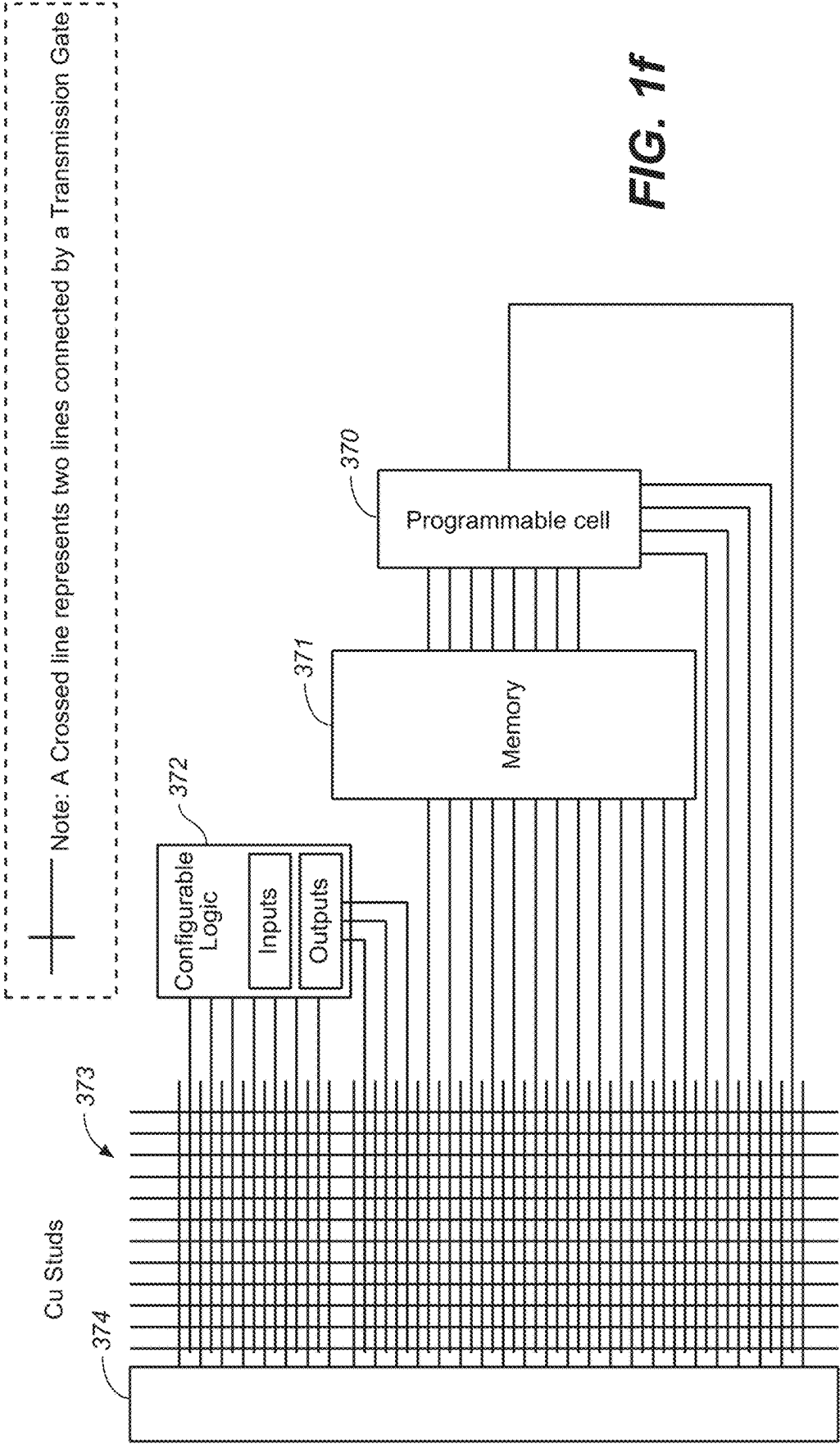
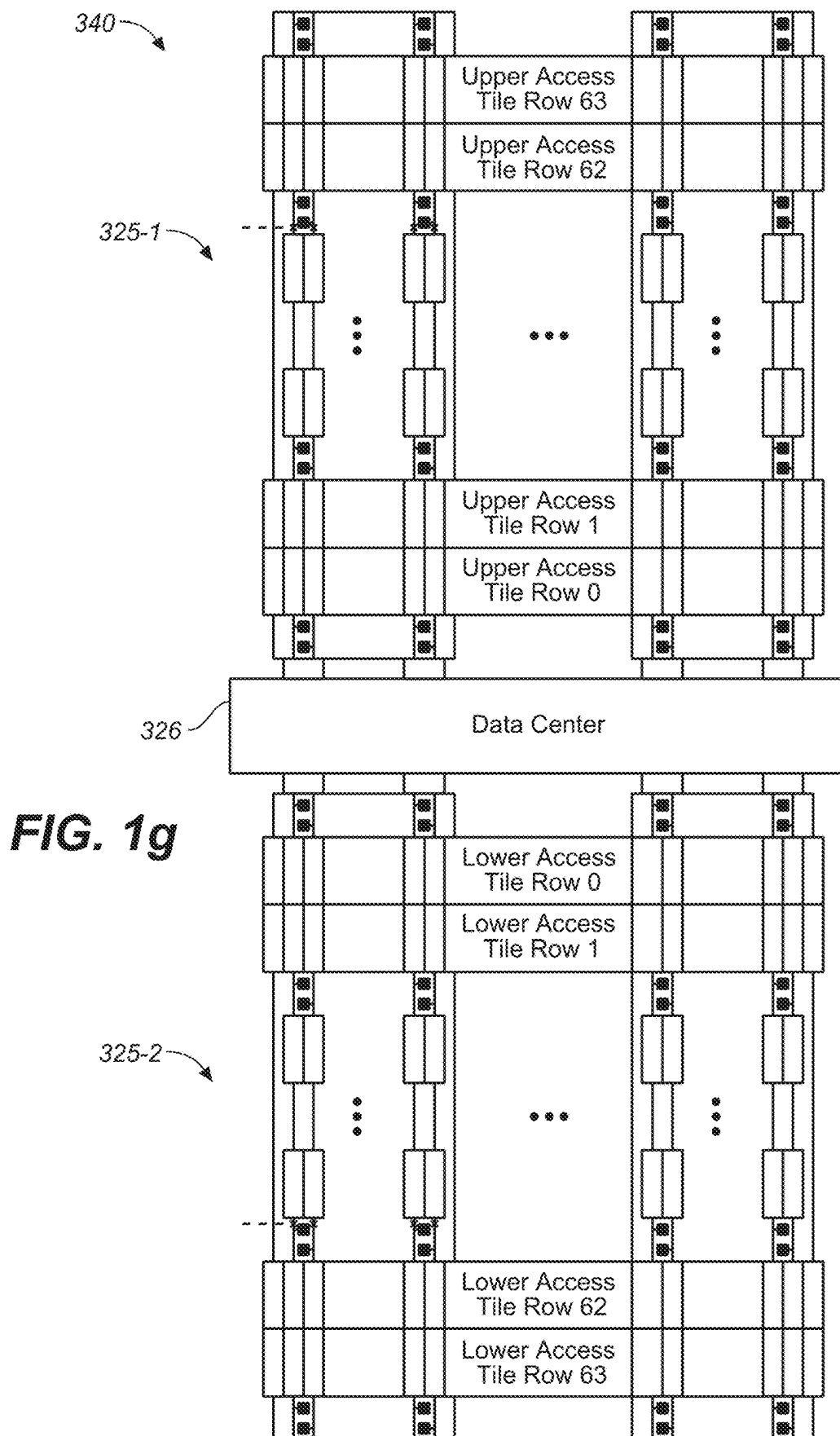


FIG. 1f



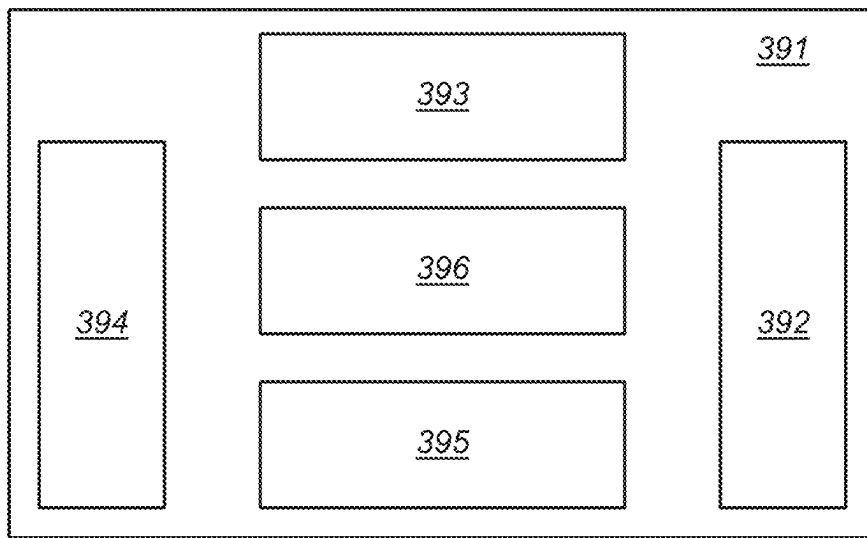


FIG. 1h

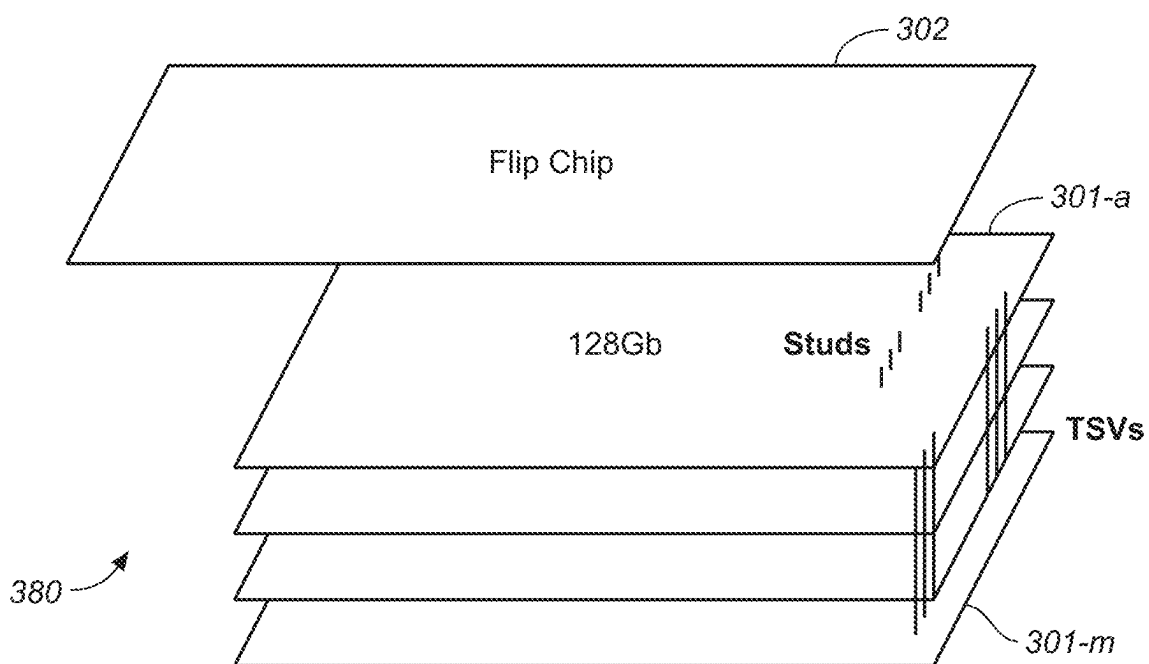


FIG. 1i

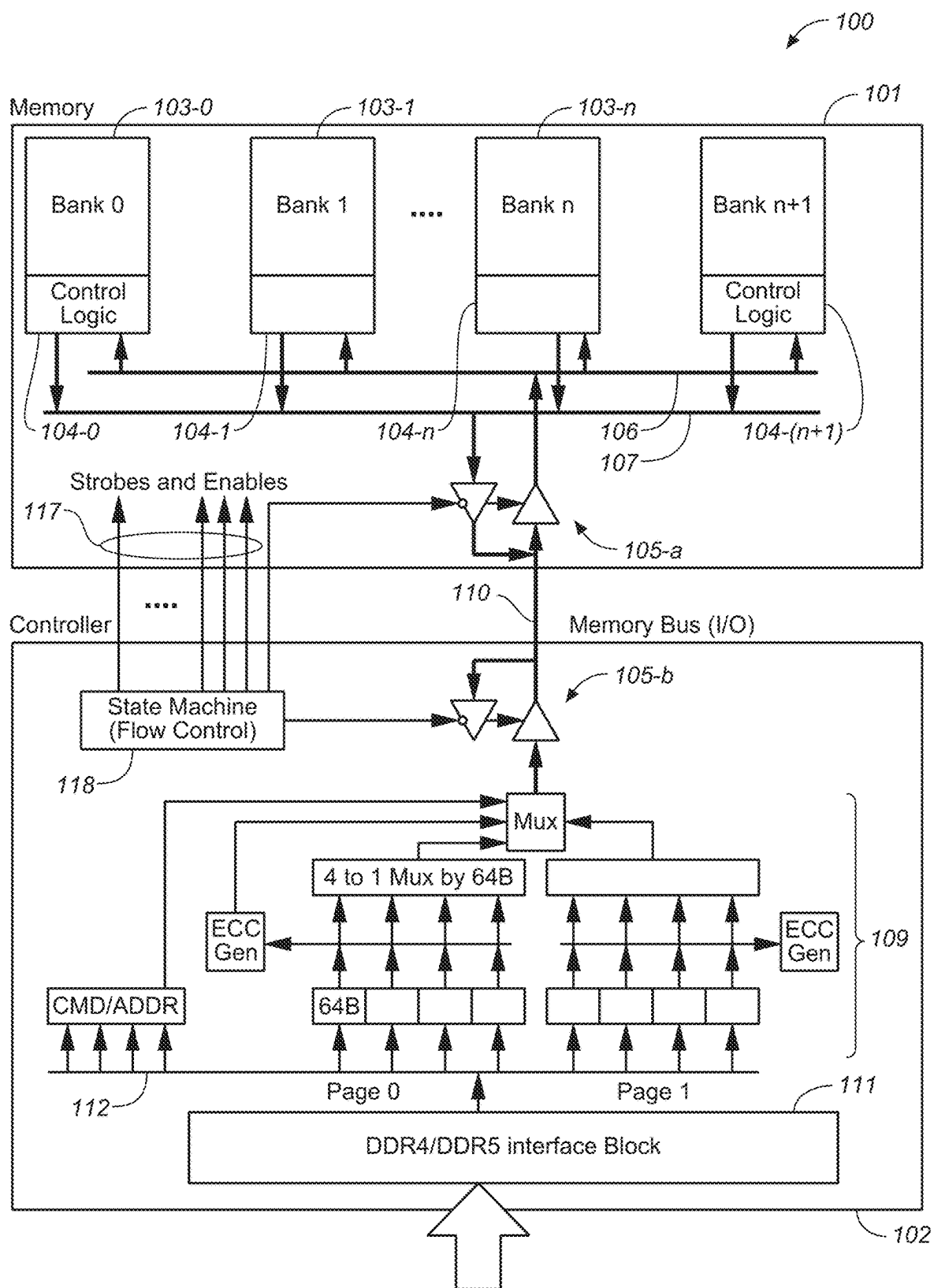


FIG. 2

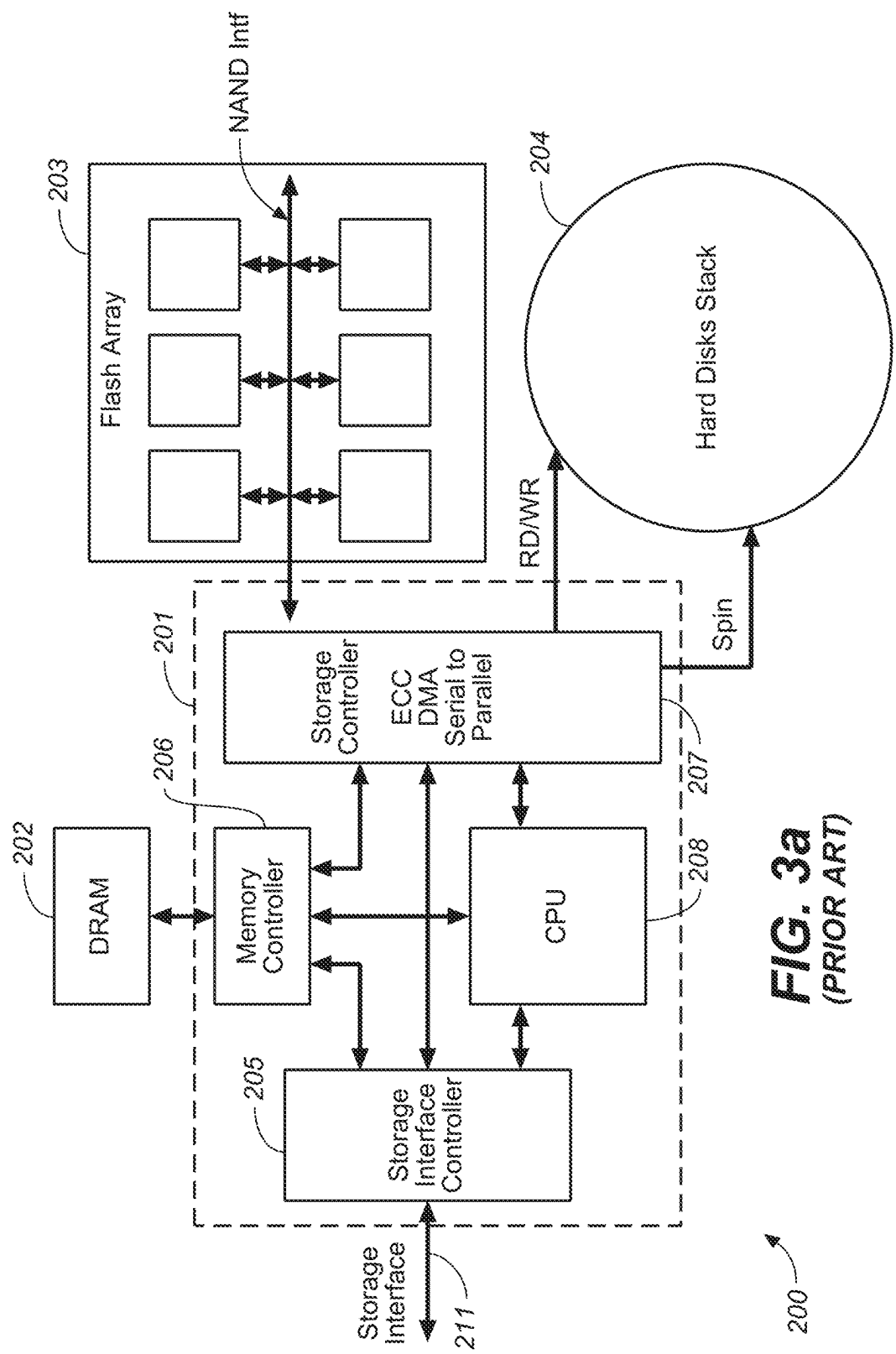


FIG. 3a
(PRIOR ART)

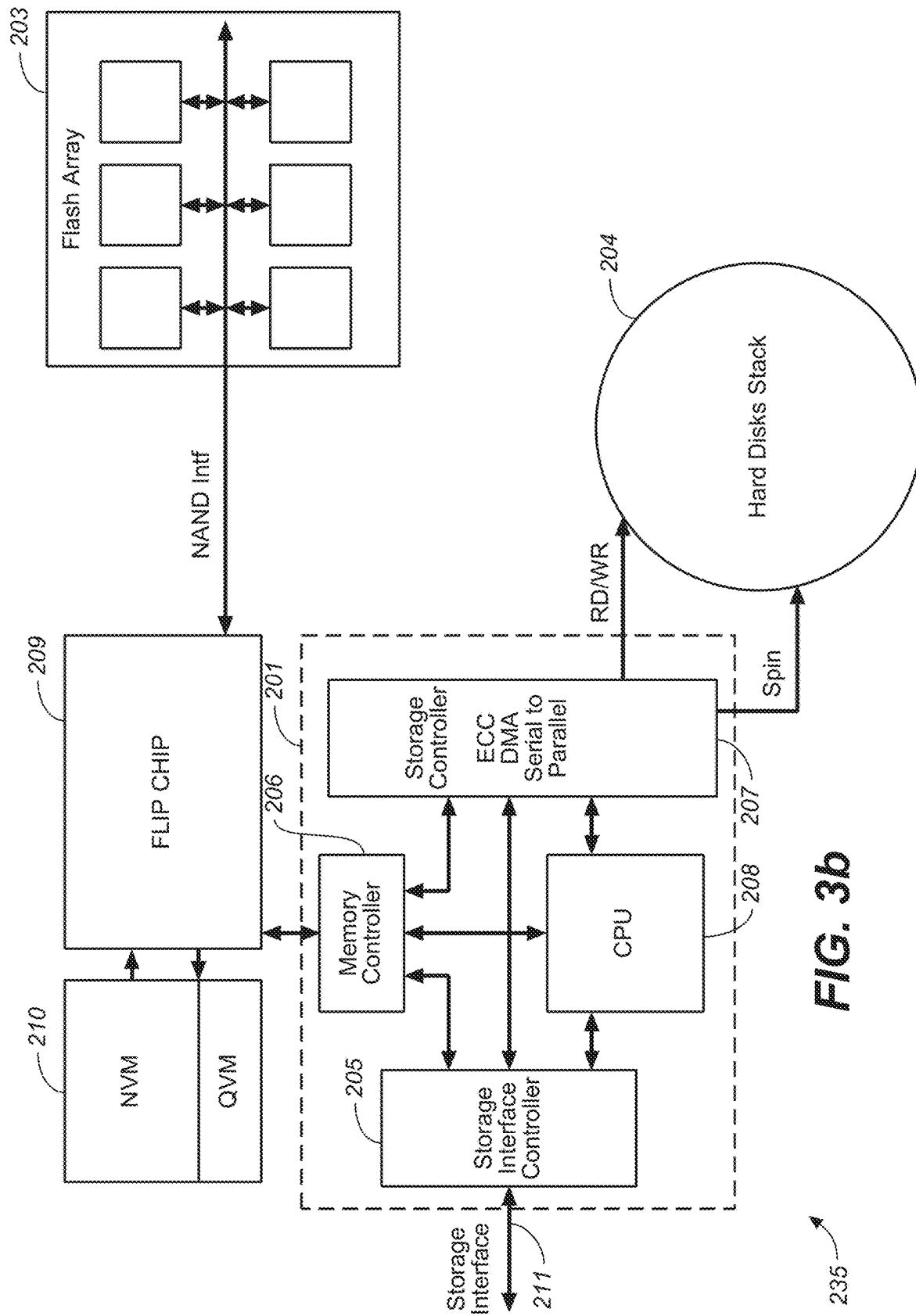


FIG. 3b

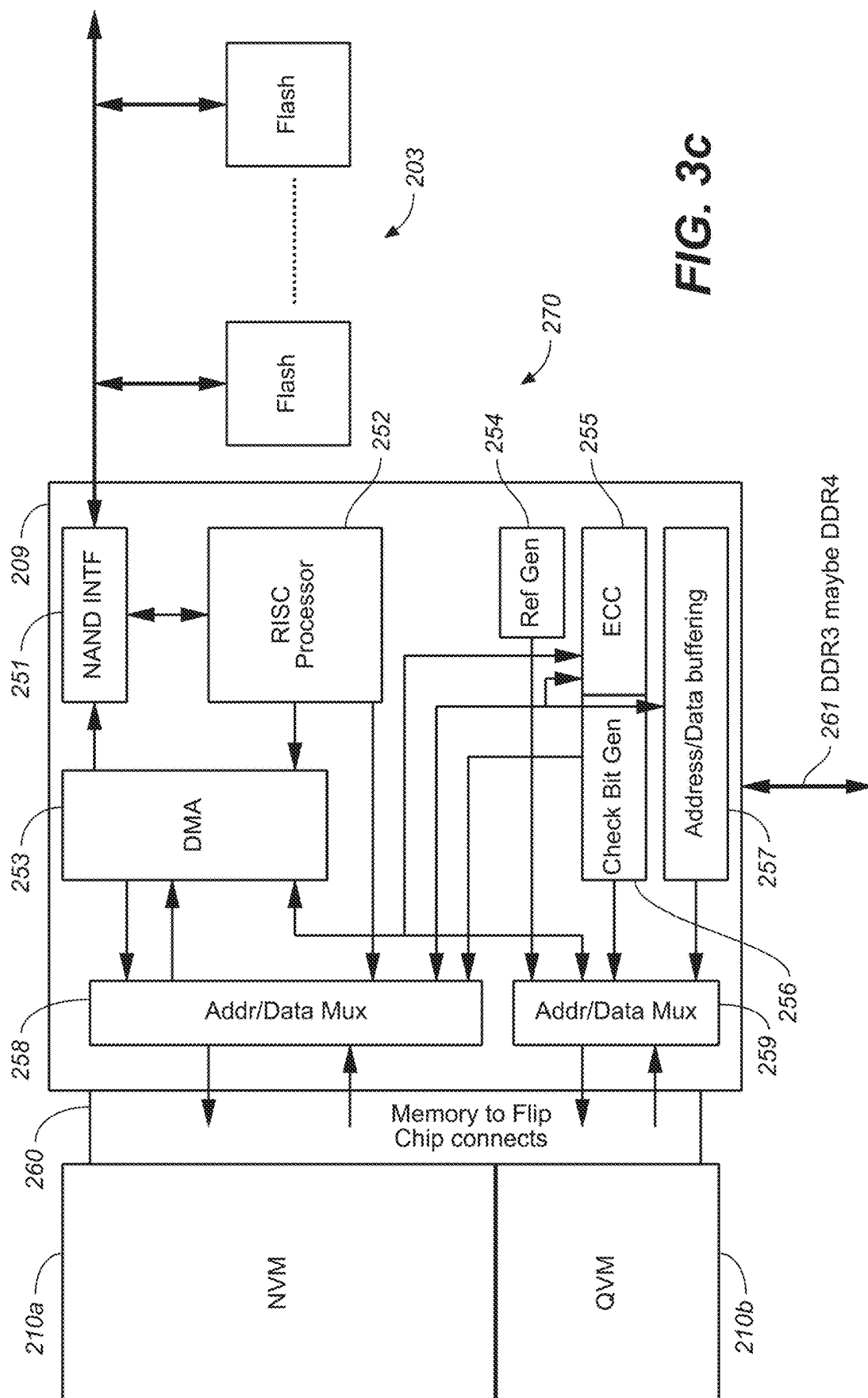


FIG. 3C

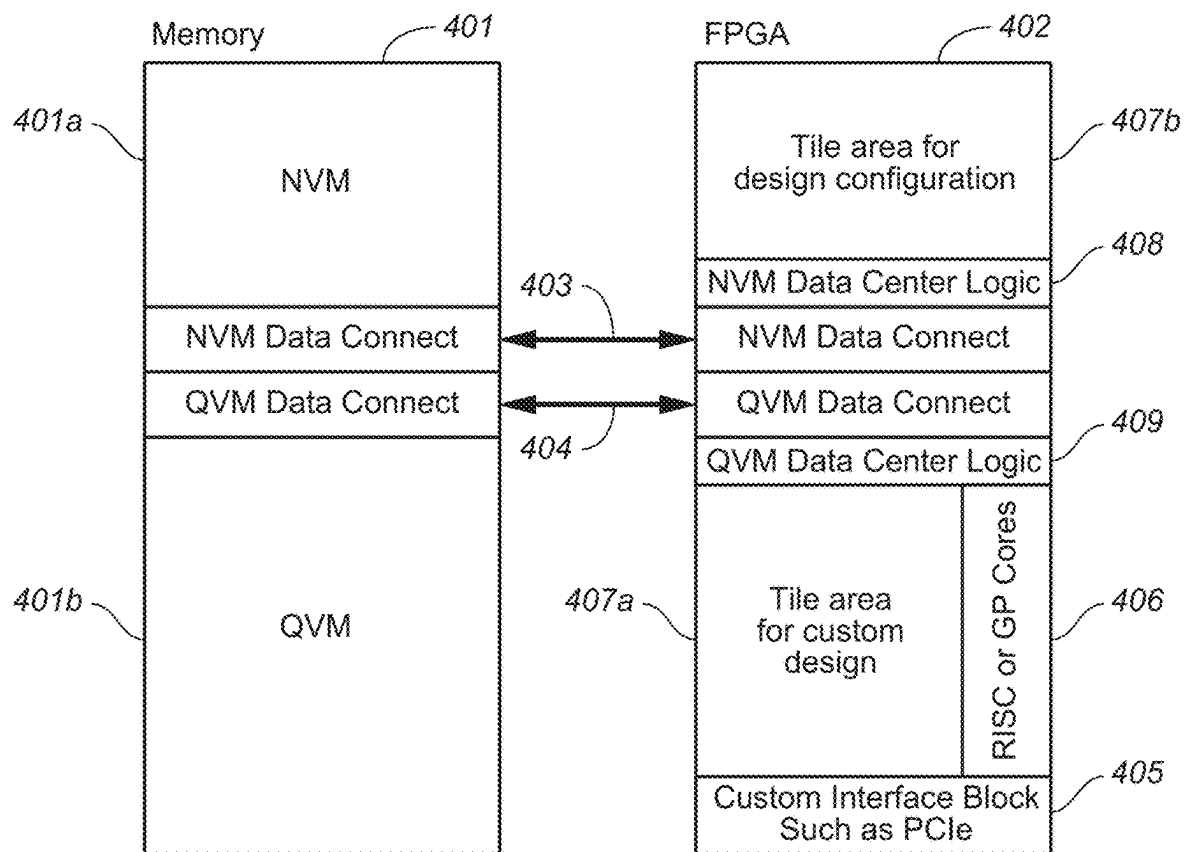
**FIG. 4a**

FIG. 4b

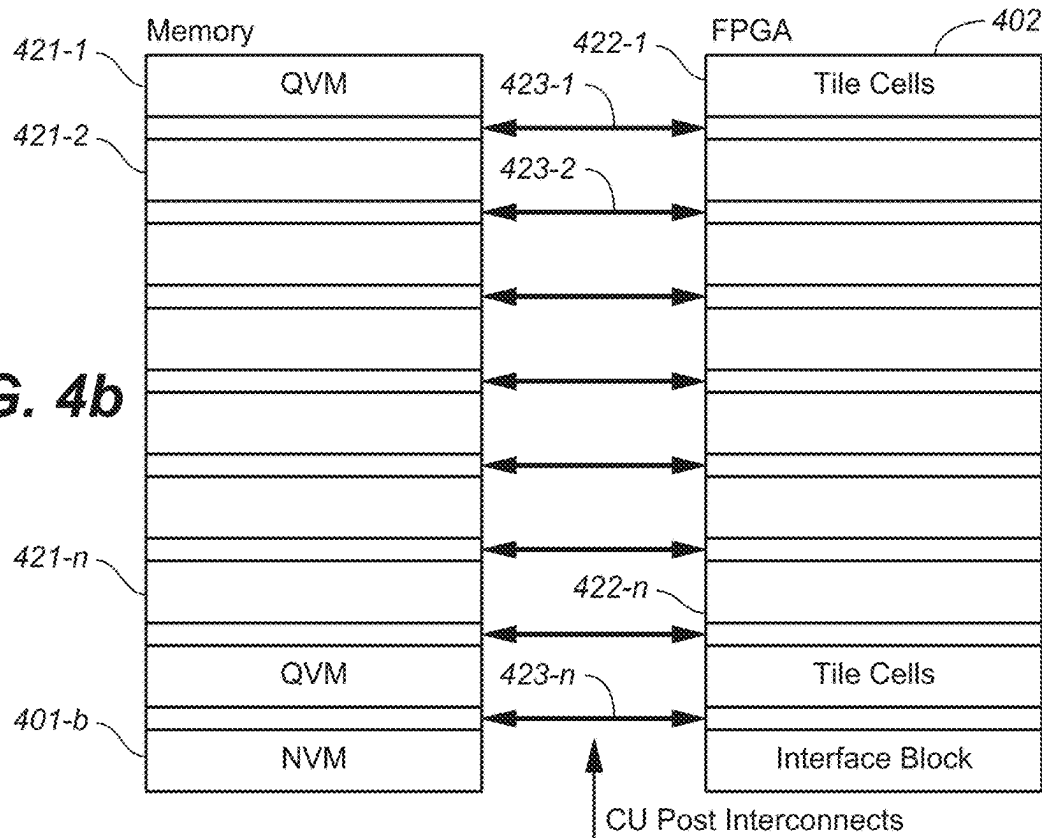
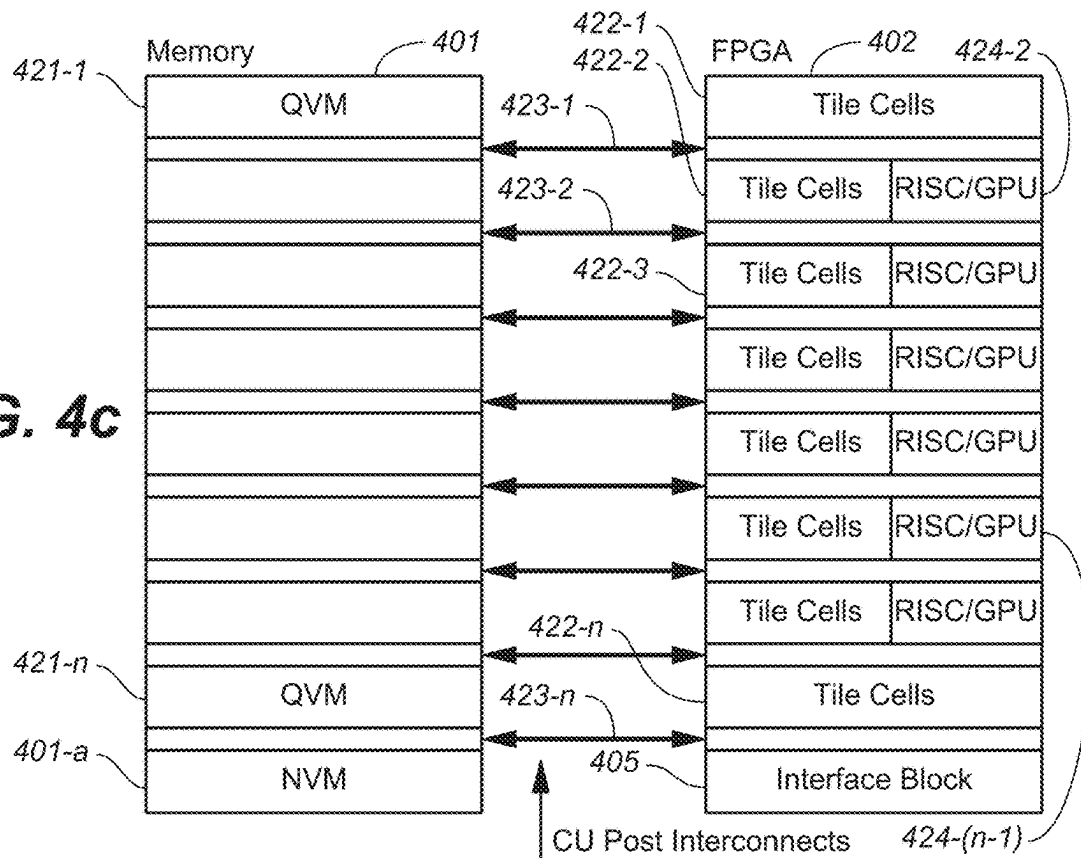
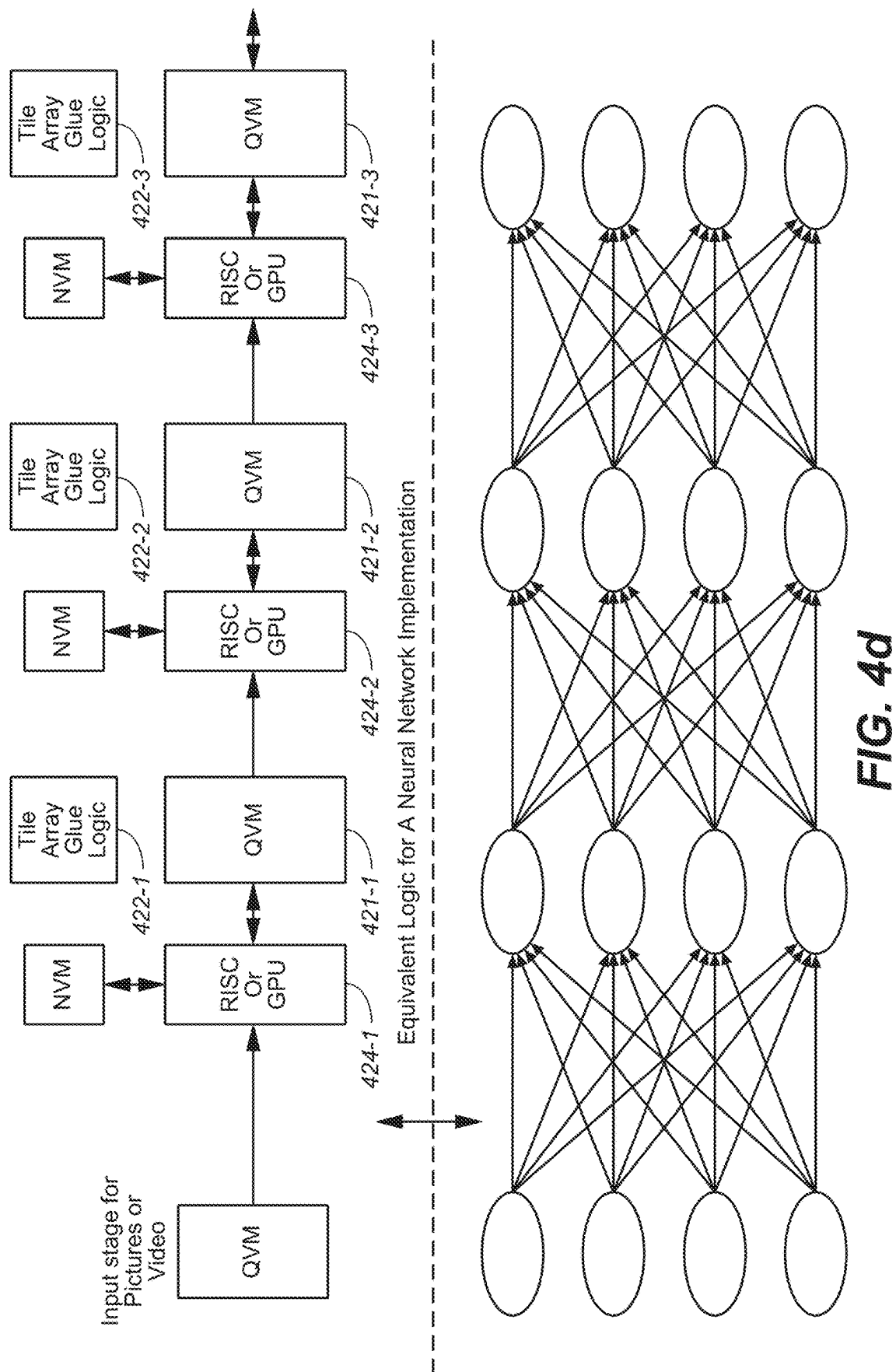
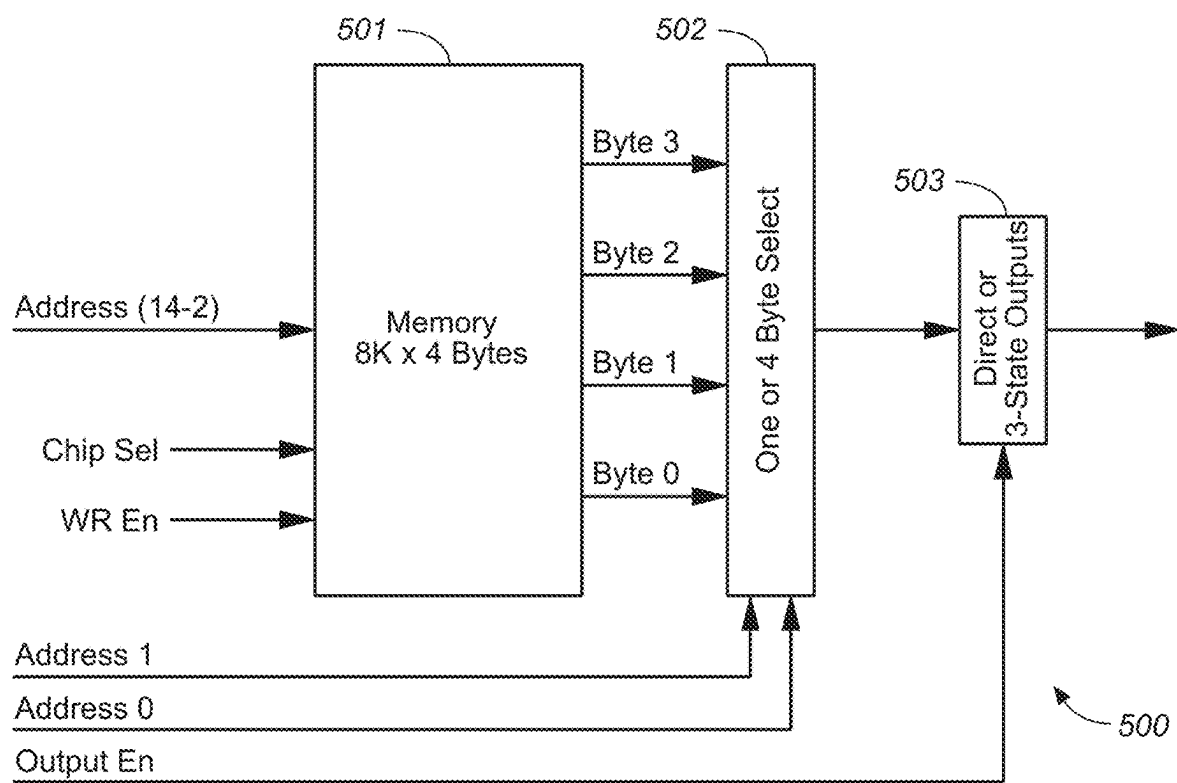


FIG. 4c





**FIG. 5a**

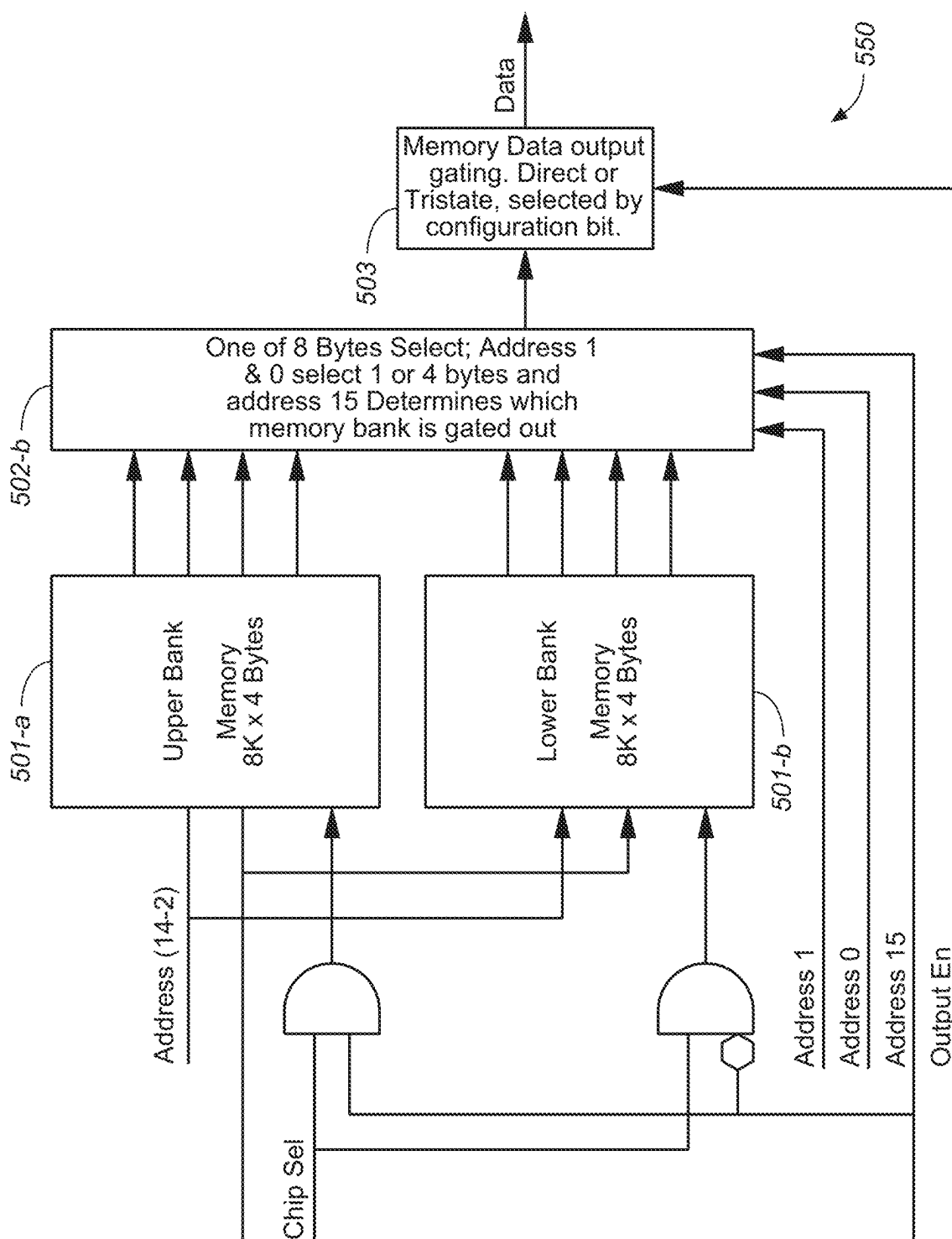


FIG. 5b

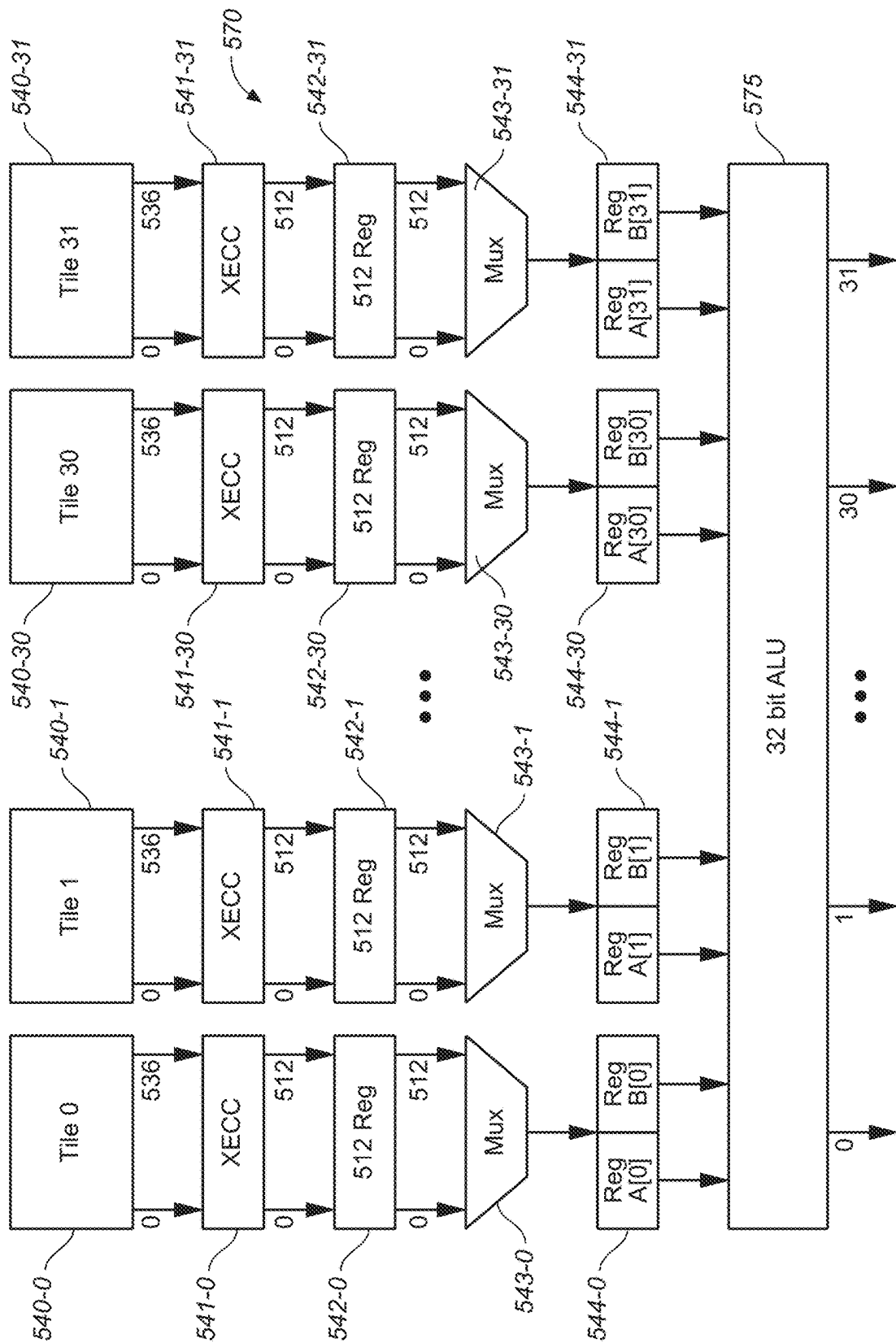
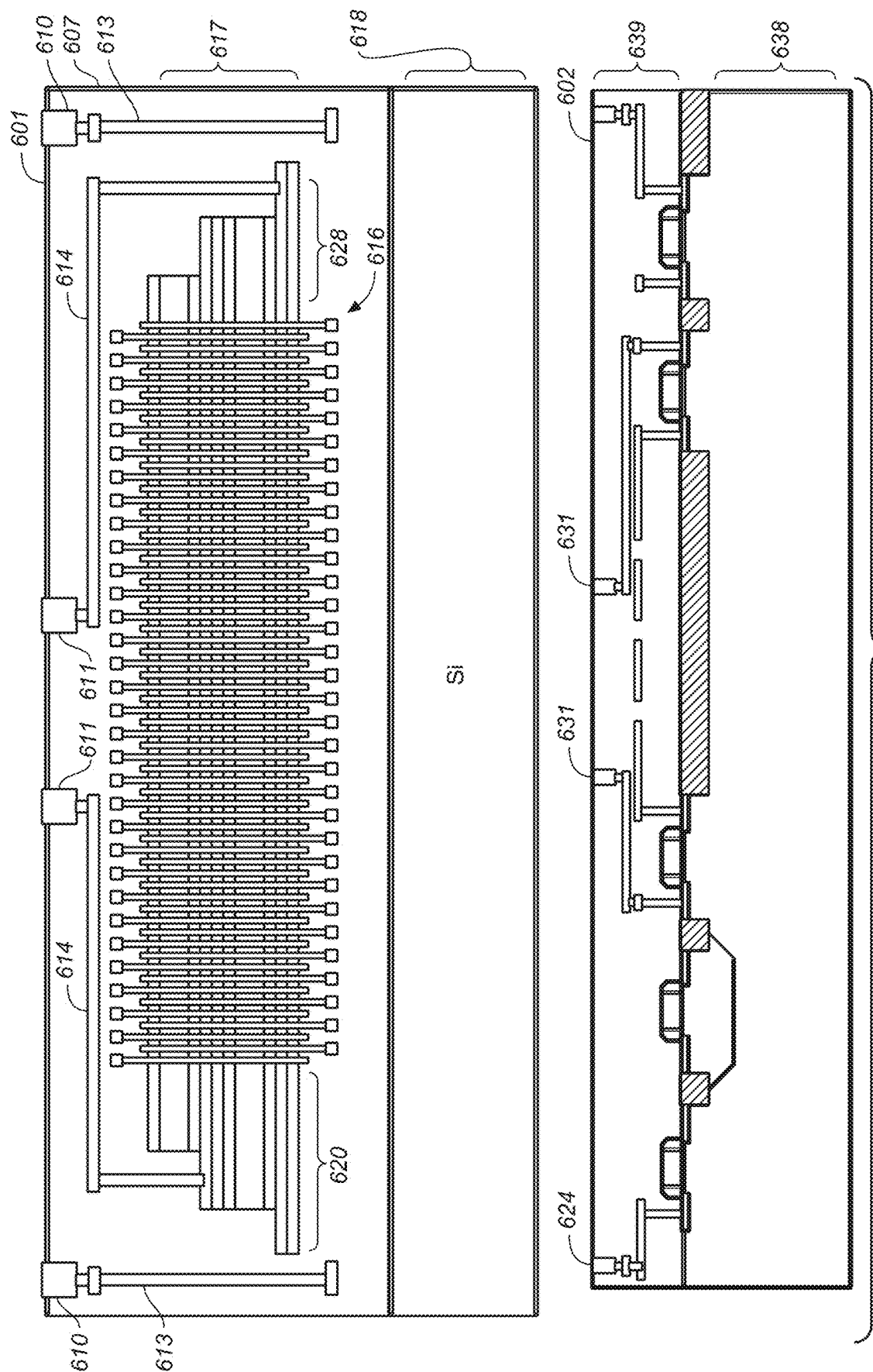


FIG. 5c



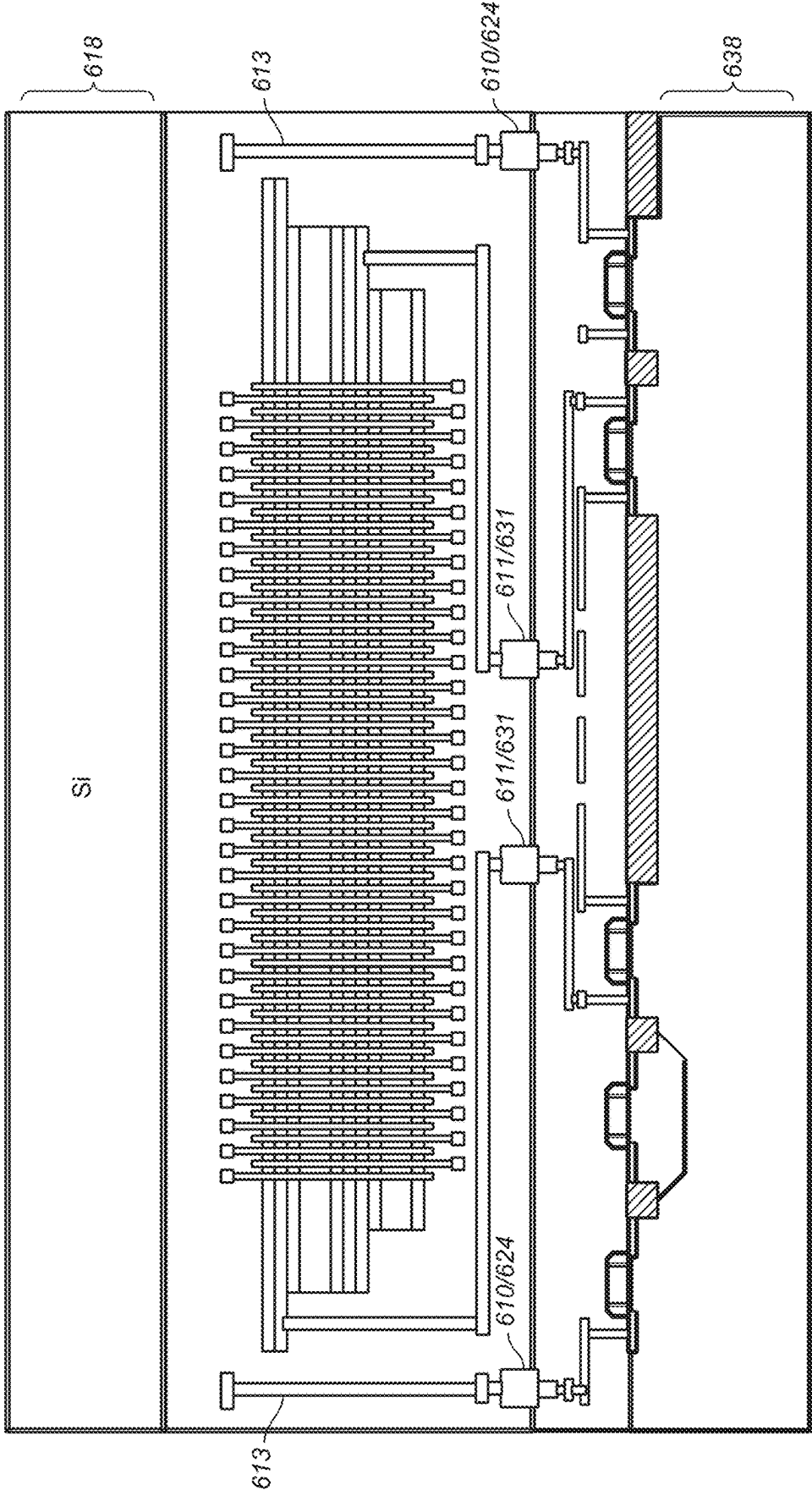


FIG. 6b

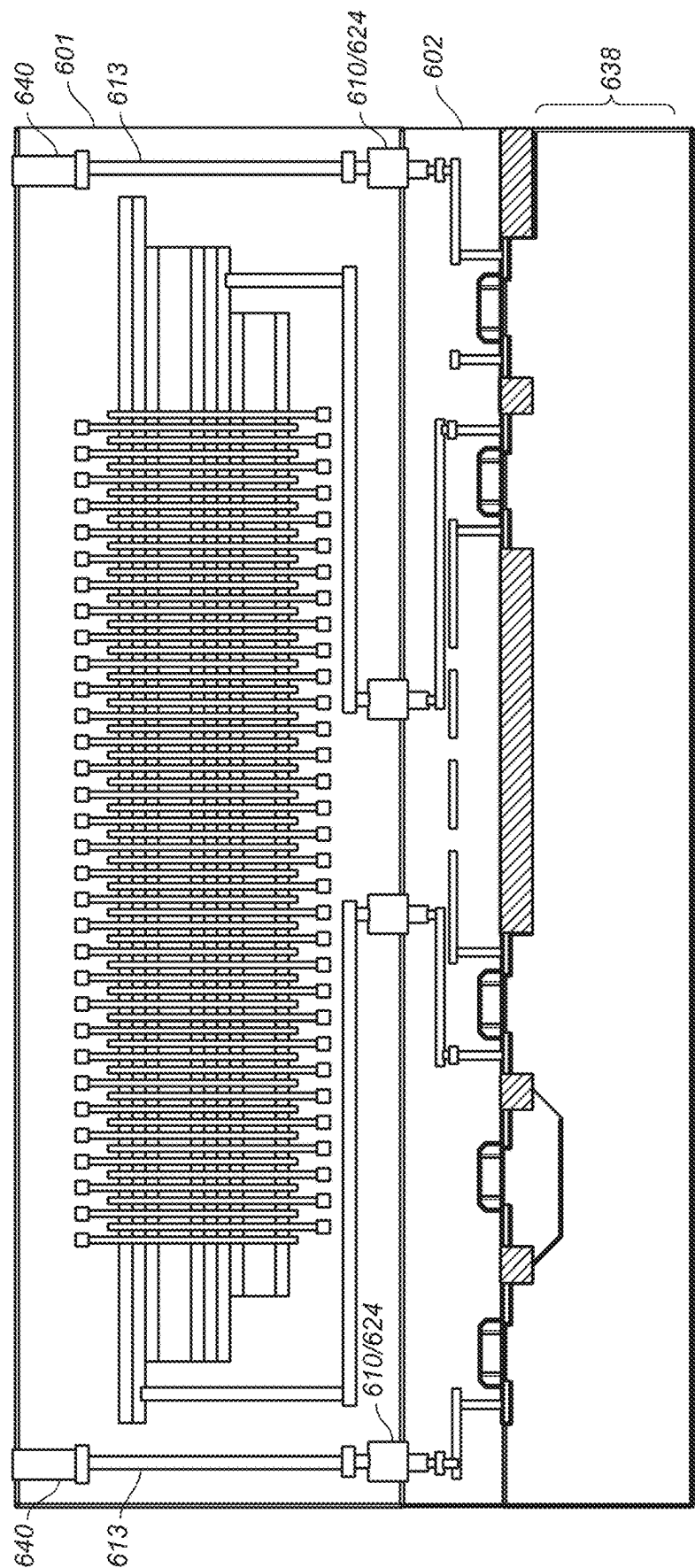


FIG. 6c

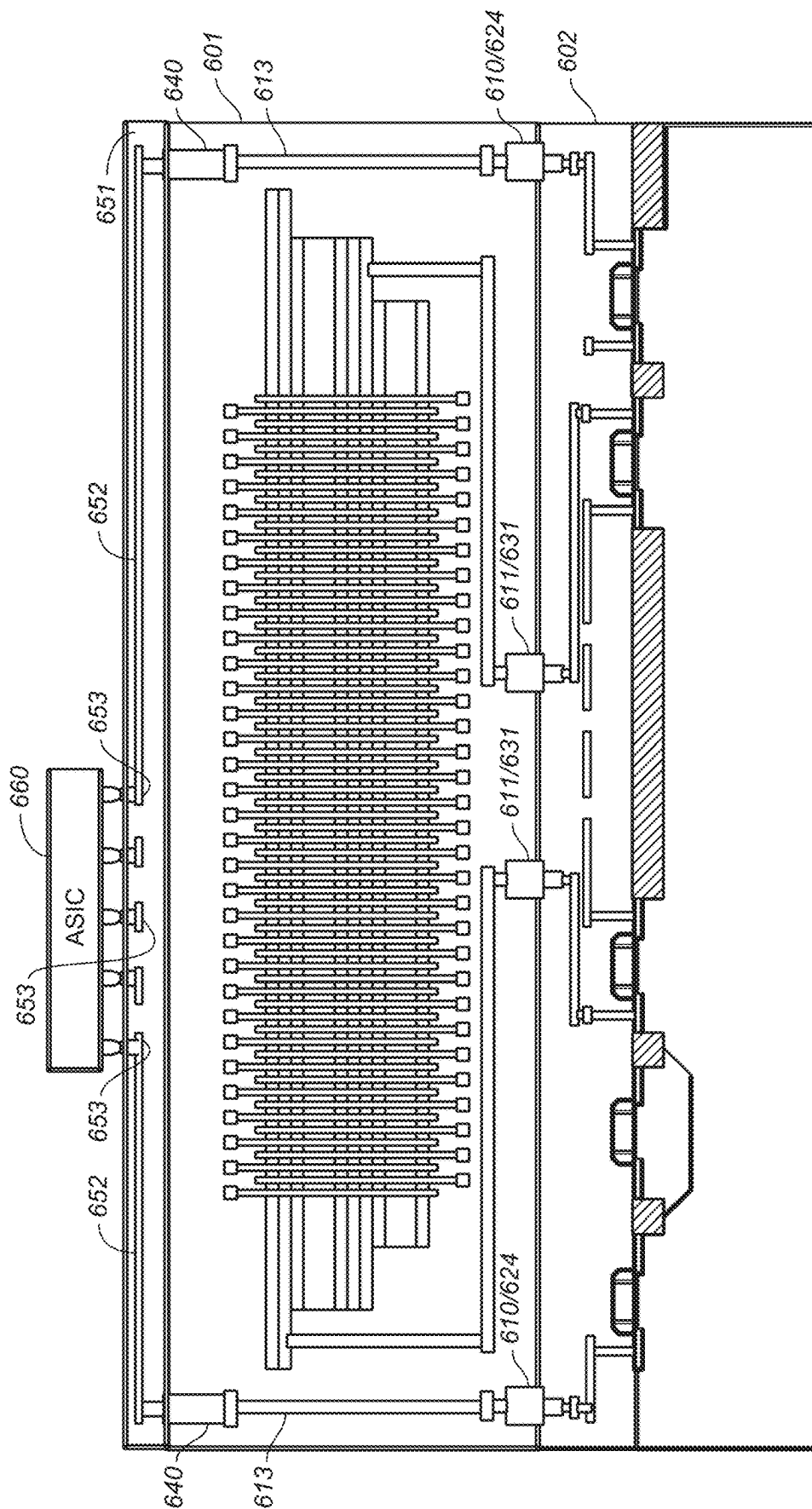
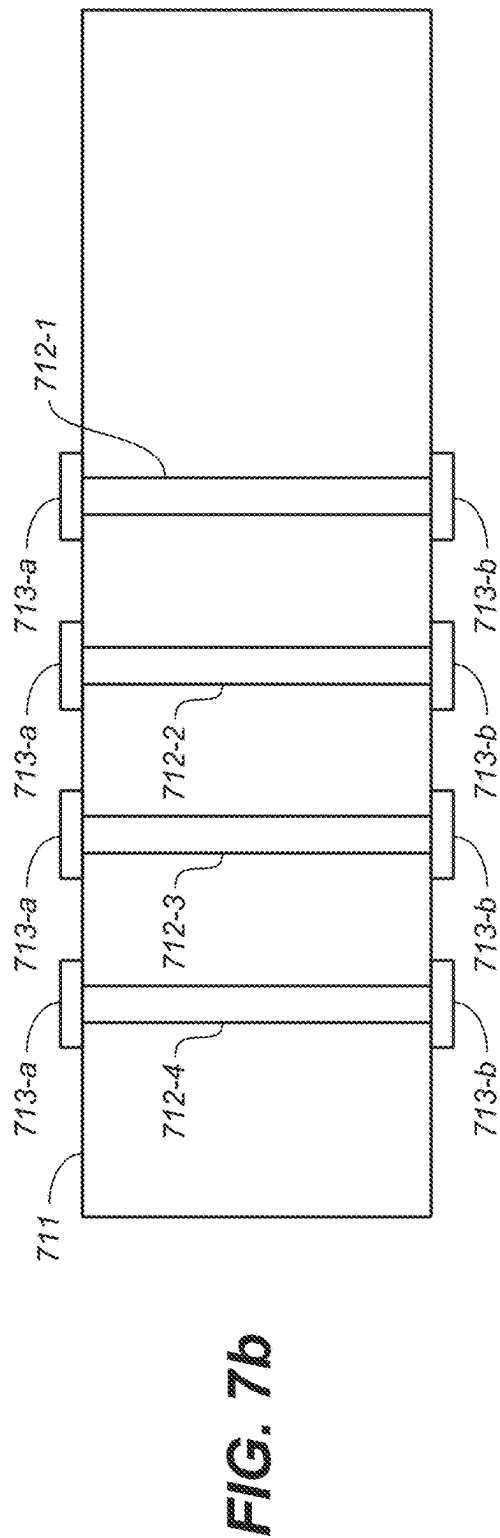
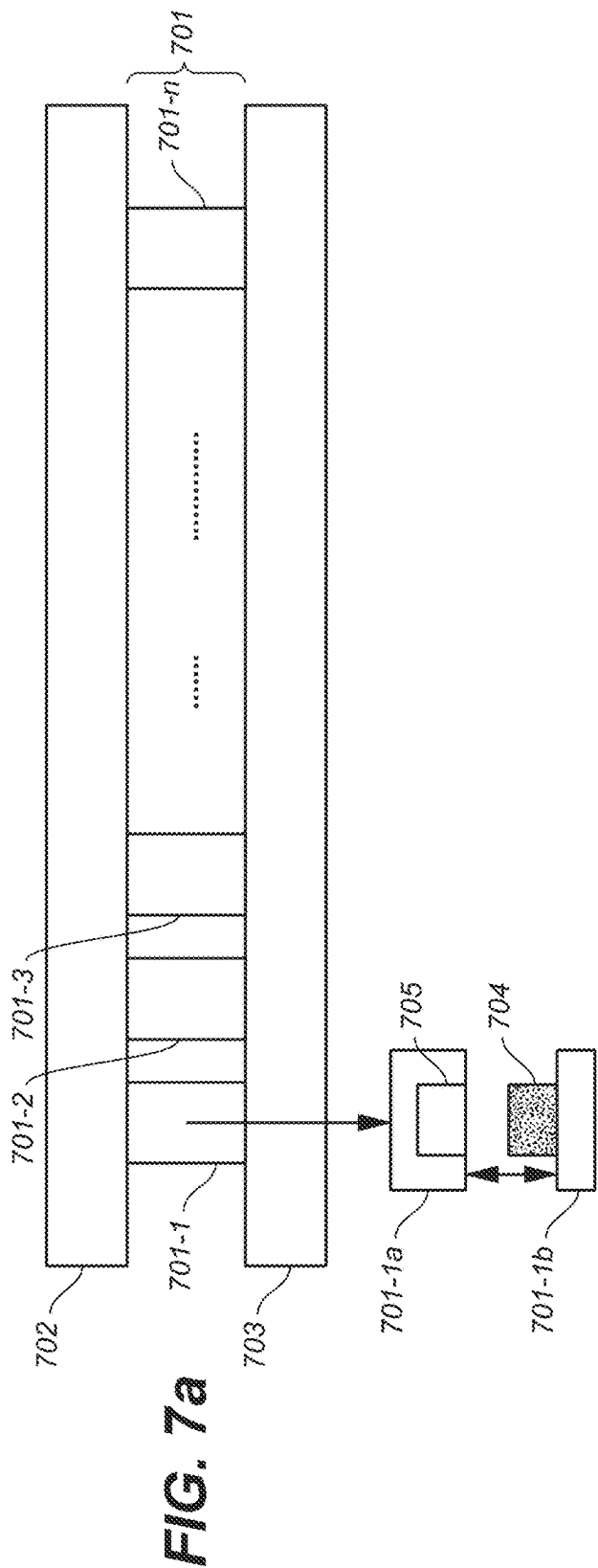
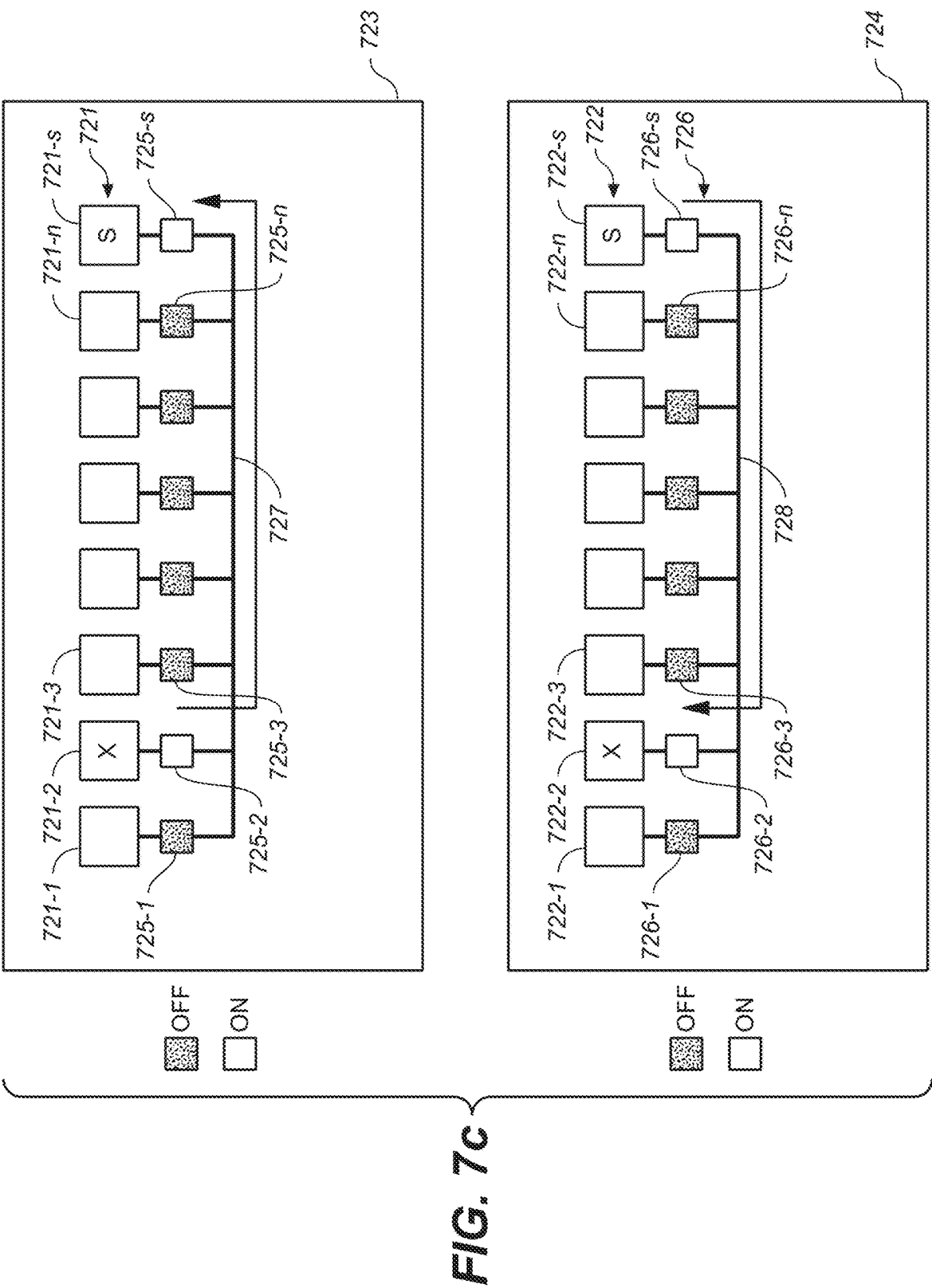


FIG. 6d





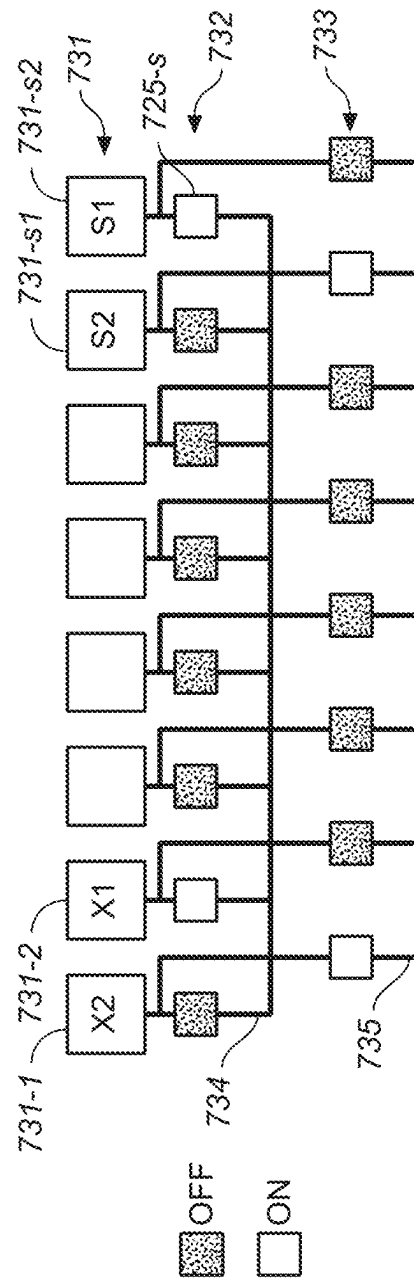


FIG. 7d

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DEVICE WITH EMBEDDED HIGH-BANDWIDTH, HIGH-CAPACITY MEMORY USING WAFER BONDING

CROSS REFERENCE TO RELATED APPLICATIONS

The present application is related to (i) U.S. patent application (“Non-Provisional Application I”), Ser. No. 16/012,731, entitled “3-Dimensional NOR Memory Array Architecture and Methods for Fabrication Thereof,” filed Jun. 19, 2018; (ii) U.S. patent application (“Non-Provisional Application II”), Ser. No. 16/107,732, entitled “Three-dimensional vertical NOR Flash Thin-Film Transistor Strings,” filed on Aug. 21, 2018; and (iii) U.S. patent application (“Non-Provisional Application III”), Ser. No. 16/579,329, entitled “Wafer Bonding in Fabrication of 3-Dimensional NOR-memory Circuits,” filed on Sep. 23, 2019, which claims priority of U.S. provisional application (“Provisional Application I”), Ser. No. 62/735,678, entitled “Wafer Bonding in Fabrication of 3-Dimensional NOR-memory Circuits,” filed on Sep. 24, 2018.

The present application is a continuation application of U.S. patent application Ser. No. 18/138,270, entitled “Device With Embedded High-Bandwidth, High-Capacity Memory Using Wafer Bonding,” filed Apr. 24, 2023, now U.S. Pat. No. 12,068,286, issued Aug. 20, 2024, which is a divisional application of U.S. patent application Ser. No. 16/776,279, entitled “Device With Embedded High-Bandwidth, High-Capacity Memory Using Wafer Bonding,” filed Jan. 29, 2020, now U.S. Pat. No. 11,670,620, issued Jun. 6, 2023, which is related to and claims priority of (i) U.S. provisional application (“Provisional Application II”), Ser. No. 62/798,673, entitled “Device with Embedded High-Bandwidth, High-Capacity Memory using Wafer Bonding,” filed on Jan. 30, 2019; (ii) U.S. provisional application (“Provisional Application III”), Ser. No. 62/803,689, entitled “Device with Embedded High-Bandwidth, High-Capacity Memory using Wafer Bonding,” filed on Feb. 11, 2019; and (iii) U.S. provisional application (“Provisional Application V”), Ser. No. 62/843,733, entitled “Device with Embedded High-Bandwidth, High-Capacity Memory using Wafer Bonding,” filed on May 6, 2019.

The present application is also related to U.S. provisional application (“Provisional Application IV”), Ser. No. 62/735,662, entitled “Epitaxial Monocrystalline Channel for Storage Transistors in 3-Dimensional Memory Structures and Methods for Formation Thereof,” filed on Sep. 24, 2018.

The disclosures of the Non-provisional Applications I and II (collectively, the “Non-provisional Applications”) and the Provisional Applications I-V (collectively, the “Provisional Applications”) are hereby incorporated by reference in their entirety.

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present invention relates to high-performance computing. In particular, the present invention relates to creating a high-performance electronic device by providing logic integrated circuit access to a high-bandwidth, high-capacity memory device using wafer bonding.

2. Discussion of the Related Art

The Non-provisional Applications disclose 3-dimensional memory structures (“3-D NOR memory arrays”) formed on

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top of a planar monocrystalline semiconductor substrate. (Collectively, the 3-D NOR memory arrays on a single semiconductor substrate is referred to, hereinafter, as a “3-D NOR memory chip”). In one example, each 3-D NOR memory array on a 3-D NOR memory chip is organized as 3-dimensional array of thin-film storage transistors, with the thin-film storage transistors along one of the directions organized as one or more NOR memory strings. In this context, the term “NOR memory string” refers to a group of thin-film storage transistors sharing common source and drain regions. In Non-provisional Application I, each NOR memory string has its thin-film storage transistors formed along a direction parallel to the planar semiconductor substrate. In Non-provisional Application II, each NOR memory string has its thin-film storage transistors formed along a direction perpendicular to the planar semiconductor substrate.

The semiconductor substrate underlying the 3-D NOR memory arrays in the Non-provisional Applications may include CMOS circuitry provided for supporting memory operations. The thin-film storage transistors of each 3-D NOR memory array may be interconnected to the underlying support circuitry by one or more layers of conductors (“global interconnect layers”) provided between the memory structure and the semiconductor substrate or above the memory structure.

State-of-the-art dynamic random-access memory (“DRAM”) arrays are typically formed at the surface of a planar semiconductor substrate. As such, the 2-dimensional silicon “real estate” on the planar semiconductor substrate must be shared between its DRAM memory arrays and their support circuitry. Both the inability to form a 3-dimensional array of memory cells and having to form support circuitry on precious silicon real estate result in DRAM arrays having a much lower density per unit area of silicon substrate than the 3-D NOR memory arrays of the Non-provisional Applications. In other words, a 3-D NOR memory chip has far higher capacity than a DRAM integrated circuit fabricated on a silicon die of comparable size.

Wafer bonding (or die-bonding) is a technique used in the manufacturing of semiconductor devices. In wafer bonding, semiconductor dies are joined, for example, by thermocompression, adhesive, anodic, or thermal techniques. Provisional Application I discloses numerous examples of interconnecting devices on two or more semiconductor dies using a “flip-chip” (or “flip-wafer”) wafer bonding technique. Specifically, Provisional Application I discloses examples in which one or more of the wafer-bonded semiconductor dies have fabricated thereon the memory structures of the Non-provisional Applications. Under the “flip-chip” technique, conductor-filled vias or conductive posts (“studs”) are exposed at the top surface of each semiconductor die to allow electrical access to the devices formed under the surface in the semiconductor die. Suitable conductors to be used as studs include, for example, copper. When two such semiconductor dies are wafer-bonded, their exposed studs come into contact with each other, thereby interconnecting devices across the wafer-bonded semiconductor dies. FIG. 7(a) illustrates one type of such stud connections. As shown in FIG. 7(a), numerous studs **701-1**, **701-2**, . . . , **701-n** (collectively, studs **701**) are provided between semiconductor dies **702** and **703**. In this implementation, studs **701** result from mating of male-female portion, as illustrated by stud **701-1**. Portion **701-1a** of stud **701-1** is formed on a surface of semiconductor die **702** and includes accessible cavity **705**. Portion **701-1b** of stud **701-1** is

formed on semiconductor die 703 and includes a protrusion 704 that fits hand-in-glove into cavity 705.

In the prior art, communication over pins between wire-bonded circuits (or between packaged circuits) is not only limited in bandwidth by the number of pins available for wire-bonding (or on the packages), driving a signal between pins across a wire-bond or between two package pins requires much power and incurs a substantial delay because of the large capacitances involved. Driving a signal across the wafer-bonded semiconductor dies over abutting studs does not have these limitations.

Besides the “flip-chip” technique, other techniques for interconnecting circuits in different wafer-bonded semiconductor dies have been developed. Under one technique, commonly referred to as the “Through-Silicon-Via” (TSV) technique. In the TSV technique, multiple conductor-filled vias are provided that extend the entire thickness of each semiconductor die, such that, when the semiconductor dies are stacked one on top of another, the conductor-filled vias abut each other to provide a network of conductors through which electrical interconnection between devices formed on different semiconductor dies are made. Under the TSV technique, because the conductors carrying signals across the semiconductor dies are aligned to allow signals to be routed between any two of the stacked semiconductor dies, the TSV are typically provided at the periphery of each stacked die, and often driven from conventional I/O pads (e.g., in a conventional DRAM bus organization). The flip-chip technique is less costly in silicon real estate and enables great flexibility and options in organizing the interfaces between the wafer-bonded dies beyond conventional bus structures. FIG. 7(b) illustrates the TSV technique using a cross section of semiconductor die 711. As shown in FIG. 7(b), one implementation includes numerous vias 712 (represented by vias 711-1, 711-2, 711-3 and 711-4) formed in semiconductor die 711 using conventional etching techniques and are thereafter filled with conductive material (e.g., tungsten). On both ends of each via and exposed to the opposite sides of semiconductor die 711 are formed bonding pads—indicated by bonding pads 713a and 713b—for connections either with circuitry formed on one of the surfaces of semiconductor die 711, or through a wafer bond to circuitry on another semiconductor die or to other external circuitry.

Under another technique, commonly referred to as the “silicon interposer” technique, two or more semiconductor dies are each wafer-bonded in a “flip-chip” fashion to a large silicon substrate (i.e., the “silicon interposer”). The silicon interposer provides a network of interconnect conductors to connect the studs of the semiconductor dies. Under the “silicon interposer” technique, the surface area on the silicon interposer that abuts the wafer-bonded semiconductor is greater than the total surface areas of its wafer-bonded semiconductor dies.

A variation of the “silicon interposer” technique, referred to as the “silicon bridge” technique. Under the “silicon bridge” technique, each semiconductor die to be wafer-bonded has its studs for interconnection of devices placed on specific locations along one or more designated edges of the semiconductor die. Studs for power and ground signals may be separately provided outside of these locations. The semiconductor dies are then placed “face-down” on a surface of a circuit board, such that their respective designated edges of interconnection studs are in close vicinity of each other. In-laid in the circuit board is a silicon substrate (i.e., the silicon bridge) which provides a network of conductor to interconnect the studs of the semiconductor dies. The semi-

conductor dies are then wafer-bonded to the silicon bridge. In this manner, unlike the interposer technique, the silicon bridge need only overlay that close vicinity of interconnect studs. Outside of the silicon bridge, the circuit board provide separate access to power and ground planes.

A “High-Bandwidth Memory” (HBM) Standard (JESD235) has been promulgated by the standard organization JEDEC. Under the HBM standard, a high-bandwidth memory device is achieved by stacking up to eight DRAM dies and, optionally, a base “logic” die with a memory controller, which are interconnected by TSV and micro-bumps. Essential features of the HBM Standard is disclosed in *Highlights of the High-Bandwidth Memory (HBM) Standard*, at the Memory Forum, Jun. 14, 2014, available from Nvidia Corporation. Under the HBM standard, the DRAM dies provide a number of completely independent data interfaces (“channels”), with each channel providing a 128-bit bus interface that is similar to a conventional DDR bus interface. HBM addresses the pin-out bottleneck by bonding a stack of memory wafers or dies to another semiconductor die (e.g., a logic circuit) using an interposer wafer using the TSV technique. Using an eight-wafer stack, HBM can increase the memory pin-out by a factor of eight (e.g., 128 or 256 output signals). Significant silicon “real estate” is required to implement the data interfaces under IBM.

SUMMARY

According to one embodiment of the present invention, an electronic device with embedded access to a high-bandwidth, high-capacity fast-access memory includes (a) a memory circuit fabricated on a first semiconductor die, wherein the memory circuit includes numerous modular memory units, each modular memory unit having (i) a three-dimensional array of storage transistors, and (ii) a group of conductors exposed to a surface of the first semiconductor die, the group of conductors being configured for communicating control, address and data signals associated with the memory unit; and (b) a logic circuit fabricated on a second semiconductor die, wherein the logic circuit also includes conductors each exposed at a surface of the second semiconductor die, wherein the first and second semiconductor dies are wafer-bonded, such that the conductors exposed at the surface of the first semiconductor die are each electrically connected to a corresponding one of the conductors exposed to the surface of the second semiconductor die. The three-dimensional array of storage transistors may be formed by NOR memory strings. The memory circuit may be, at least in part, a quasi-volatile memory circuit having an endurance capable of a million or more write-erase cycles. The wafer bonding may be achieved preferably using a flip-chip or flip-wafer technique; alternatively, other wafer-bonding techniques, such as TSV, silicon interposer or silicon bridge techniques, may be used in lieu of or in conjunction with the flip-chip technique.

According to one embodiment of the present invention, the modular memory units are formed above a planar substrate of the first semiconductor die and placed in a regular configuration. The regular configuration may arrange the memory units along rows and columns, such that the modular memory units may be configured according to a memory segmentation scheme into memory segments that are independently addressable (a) by memory unit individually, (b) row-by-row, or (c) block-by-block, wherein each block of memory units consists of memory units within a predetermined number of rows and a predetermined number of columns. Memory segmentation may be achieved using

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configuration cells, which stored values configure signal paths for connecting the control, address and data signals of the memory units to their respective groups of conductors according to the memory segmentation scheme. Alternatively, anti-fuses may be used to set the configuration. The signal paths may be implemented by a network of switches (e.g., transmission gates) interconnecting a network of conductors. The configuration cells may be made field-programmable.

According to one embodiment of the present invention, the modular memory units includes a data processing circuit that processes data (e.g., error correction and check-bit generation) to be stored into or read from the modular units. In one implementation, the modular memory units are placed on opposite sides of the data processing circuit. In one embodiment, the modular memory units are assigned to different memory segments, with each memory segment being provided a separate portion of the data processing circuit for data processing.

In one embodiment, the memory circuit includes a quasi-volatile memory (QVM) circuit. In another embodiment, the memory circuit may include both QVM circuitry and non-volatile memory (NVM) circuitry on the same semiconductor die. The QVM of the present invention has short read, erase and write latencies, preferably comparable or approaching those of DRAM, and an erase-write cycle endurance that is one or more orders of magnitude greater than conventional NAND flash memory or 3-D NAND flash memory.

According to one embodiment of the present invention, a data processing circuit in the logic circuit provides data processing (e.g., error correction and check-bit generation) for data read from or to be stored into the memory circuit. The logic circuit may include custom logic circuits, such as microprocessors (e.g., RISC-type processor or graphics processing units). In addition, the logic circuit may be provided one or more of: industry standard data interfaces, and field programmable logic devices.

According to one embodiment, both the memory circuit and the logic circuit may be segmented, and their resources paired by segments to allow parallel computing operations. Such an organization provides great advantage in some applications, such as multi-processor system (e.g., multiple core CPUs or GPUs) on the logic circuit, with each processor being paired with one or more corresponding memory segments in the memory circuit, neural networks, as well as other artificial intelligence-related circuitry. These segments may also be organized as a data pipeline to implement a sequence of related operations each receiving as input data resulting from a previous operation and temporarily stored on its memory segment.

According to one embodiment of the present invention, the modular memory arrays may be used as programmable logic circuits implemented as look-up tables,

According to one embodiment of the present invention, the electronic device may implement a storage system controller circuit, which includes (i) a storage controller for managing a storage system (e.g., a hard disk system or a NAND flash storage system); and (ii) a flash controller for managing a flash cache memory for the storage system, wherein the flash controller includes a memory circuit wafer-bonded to a logic circuit. In one embodiment, the logic circuit in the flash controller includes a memory controller for the memory circuit, which may include QVM and NVM circuits. The memory controller in the logic circuit may have an industry standard data interface, such as a DRAM interface, so that the memory controller may be

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accessed in the same manner as a DRAM controller. The industry standard bus interface may also be a PCI/e interface. The memory controller may further implement an interface to a NAND flash memory circuit, to allow the NAND flash memory to interact with the QVM cache memory for the storage system.

The present invention is better understood upon consideration of the detailed description below, in conjunction with the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1a shows a floor plan of semiconductor die 360 that includes memory circuit 355 made up by arrays 355a and 355b of tiles, suitable for use in one embodiment of the present invention.

FIG. 1b shows system 300, which includes QVM circuit 301 that is wafer-bonded to processor circuit 302 using a flip-chip or flip-wafer technique, according to one embodiment of the present invention.

FIG. 1c shows memory circuit 320 which includes 64 rows by 32 columns "core" of tiles, suitable for implementing a portion of QVM circuit 301, in accordance with one embodiment of the present invention.

FIG. 1d is a magnified view of a portion of memory circuit 320, in which adjacent studs 324-a-n and 324-(a+1)-(n+1) provided to adjacent local bit lines for bits n and (n+1) between two tile-pairs.

FIG. 1e illustrates tile-by-tile studs programmability at the memory tile-level and data path selection at the data center-level, in accordance with one embodiment of the present invention.

FIG. 1f is a schematic diagram illustrating configuration of studs, according to one embodiment of the present invention.

FIG. 1g shows memory circuit 340, which includes cores 325-1 and 325-2 that share data center 326, suitable for implementing a portion of QVM circuit 301, in accordance with one embodiment of the present invention.

FIG. 1h shows a multi-die configuration implemented using the "flip-chip" technique, in conjunction with a silicon interposer, according to one embodiment of the present invention.

FIG. 1i shows system 380 in which multiple memory circuits 301a to 301-m are wafer-bonded to processor circuit 302 using both the flip-chip and the TSV techniques.

FIG. 2 is a block diagram of memory system 100, which includes memory circuit 101 formed on a semiconductor substrate that is wafer-bonded to controller circuit 102 formed on a second semiconductor substrate, according to one embodiment of the present invention.

FIG. 3a shows storage system 200 in the prior art for a storage system that includes storage device 204 with flash cache memory 203.

FIG. 3b shows storage system 200 that includes NVM and QVM circuits 210, which are wafer-bonded to multi-interface memory controller circuit 209, according to one embodiment of the present invention.

FIG. 3c shows one implementation 270 of NVM and QVM circuits 210 and multi-interface controller circuit 209, according one embodiment of the present invention.

FIG. 4a shows system 400, in which high-capacity memory circuit 401 is wafer-bonded using the flip-chip technique to FPGA circuit 402, according to one embodiment of the present invention.

FIG. 4b is a logical representation of a configuration in which QVM portion 401b of memory circuit 401 is seg-

mented into memory segments **421-1**, **421-2**, . . . , and **421-n**, respectively serving programmable logic (“tile cells”) segments **422-1**, **422-2**, . . . , and **422-n** over data buses **423-1**, **423-2**, . . . , and **423-n**.

FIG. **4c** a logical representation of a variation of the configuration of FIG. **4b**, in which tile cell segments **422-1**, **422-2**, . . . , and **422-n** is also paired with custom logic circuits.

FIG. **4d** illustrates a neural network application implemented by the configuration of FIG. **4c**, in which each memory segment and its associated tile cell segment and custom logic implement a stage of a neural network, in accordance with one embodiment of the present invention.

FIG. **5a** shows programmable logic tile cell **500** implemented in **8K×4** byte memory array **501**, which can provide an 8-bit output datum directly to an embedded processor or similar logic circuit.

FIG. **5b** shows programmable logic tile cells **501a** and **501b** in a memory circuit configured to provide to make a **16K×4**-byte memory array.

FIG. **5c** shows modular memory segment **570** that has associated ALU **575** for arithmetic and logic operations, in accordance with one embodiment of the present invention.

FIGS. **6a**, **6b**, **6c** and **6d** illustrate the use of through vias in a QVM circuit that is wafer-bonded with two other circuits in accordance with one embodiment of the present invention.

FIG. **7a** illustrates one manner of providing flip-chip stud connections between two bonded wafers.

FIG. **7b** illustrates the TSV technique using a cross section of semiconductor die **711**.

FIG. **7c** illustrates a routing scheme that allows recovery from a failure in interconnection studs or a TSV.

FIG. **7d** shows an expanded routing scheme from the routing scheme of FIG. **7c** that provides recovery for any two failures in each row of bonding pads by providing an additional row of transmission gates and an additional common conductor.

For clarity of presentation and to allow cross referencing among the figures, like elements in the figures are assigned like reference numerals.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

A powerful electronic device of the present invention is formed by combining a memory circuit fabricated on one semiconductor die (e.g., a 3-D NOR memory chip) with a complex logic circuit (e.g., a memory controller, one or more multi-core processor, a field programmable logic circuit, or a neural network) formed on a second semiconductor die using a wafer-bonding or die-bonding technique. Preferably, the memory circuit comprises one or more regular arrays of addressable modular structures or building blocks of memory cells (“tiles”) placed in a regular manner. The modularity allows the memory circuit to be segmented into independently addressable memory segments. For example, a memory segment of a desired size (e.g., a row of 32 tiles) may be achieved by configuring a group of tiles to form the memory segment, as desired.

The present invention is particularly advantageous when the memory circuit is one of the high-capacity and fast-access memory circuits disclosed in the Non-Provisional Applications. Some of the memory arrays disclosed therein may be configured as non-volatile memory (NVM) circuits with a long data-retention time (e.g., tens of years). The Non-provisional Applications also disclose examples of

quasi-volatile memory (QVM) circuits that have a shorter data-retention time (e.g., up to tens of minutes), but faster access time (e.g., less than 100 nanoseconds). Because of their fast access times, such QVM memory circuits may be used as run-time memory, comparable to DRAMs. The NVM and QVM of the Non-provisional Applications may be organized as NOR memory strings which contribute to a read data-access time that is significantly faster than conventional NAND flash memory strings. For example, the NVM and QVM disclosed in Non-provisional Application I may be read in approximately 100 ns, compared to 50 microseconds for a NAND flash array. Furthermore, while a conventional NVM memory cell may have an endurance of less than 100,000 write-erase cycles, a thin-film storage transistor of a QVM circuit of the Non-provisional Applications have an endurance in excess of 10^9 - 10^{11} write-erase cycles, providing high tolerance to wear-out degradation. QVM is thus more suitable than NVM for memory caching applications where a high erase/write cycle count can quickly exceed the relatively low endurance limit of NVM.

When used as run-time memory, a QVM circuit requires much less frequent refreshes than a DRAM circuit. As 3-D NOR memory arrays, the QVM circuits have a higher capacity and a lower cost than DRAMs. With their fast-access and high endurance, QVMs are thus more suitable than NVMs for memory caching applications where a high erase/write cycle count can quickly exceed the relatively low endurance limit of NVM. It is possible to have both QVM and NVM memory arrays configured on the same memory chip. Also, such NVM and QVM circuits may each be multi-state (i.e., storing more than one data bit may be represented in each memory cell).

A QVM circuit, as discussed herein, is a dynamic memory requiring refresh. Compared to DRAM, however, the QVM circuits of the present invention have very small leakage of the stored charge, so that the required refresh rate is much less than that of DRAMs of comparable capacity, thereby the QVM circuits are lower power.

The advantages of a memory circuit disclosed in the Non-Provisional Applications include both high-capacity and fast access. In some of the embodiments therein, such a memory circuit can be used as non-volatile memory (NVM) because of the long data retention time (e.g., tens of years); in other embodiments, some of the memory (“quasi-volatile memory” or QVM) can be used as run-time memory—similar to DRAM—because of its fast access times. The NVM and QVM of the current invention may be constructed as three-dimensional NOR memory strings of thin-film storage transistor strings, which provide a read data access time that is significantly faster than conventional NAND flash memory arrays. For example, the NVM and QVM disclosed in Non-provisional Application I may be read in approximately 100 ns, compared to 50 microseconds for a NAND flash array.

Compared to DRAMs, QVMs leak significantly less of their stored charge, so that QVMs require a less frequent refresh rate than DRAMs, and thus QVMs operate with significantly lower power than DRAMs. While conventional DRAMs are refreshed at a millisecond-range (e.g., 64 ms under DDR2), QVMs may require refresh at a minute-range (e.g., every 10 minutes). By virtue of their three-dimensional organization (i.e., stacks of memory arrays), as illustrated, for example, in Non-provisional Application I, the QVM circuits have a higher capacity and a lower cost than DRAMs.

Using the flip-chip or flip-wafer technique, signals may be driven across the wafer-bonded semiconductor dies over the

conductive posts or studs that connect between the memory circuitry in one semiconductor die and the logic circuitry of the other semiconductor die. As the connection through the studs are relatively low in capacitance, these connections are low-power and low-latency. Without the constraint of conventional input/output circuitry, a large number of studs (e.g., at least tens of thousands) may be provided over the surface of each semiconductor die, distributed substantially uniformly over the wafer-bonded surface area. The interface under the present invention between the memory circuit and the logic circuit circumvents the package pin-limitations of the prior art, allowing potentially tens of thousands of bits or more to be transferred simultaneously across the semiconductor dies. Hence, an electronic device of the present invention has the advantages of a large embedded high-bandwidth interface, much like an internal data highway with tens of thousands or more lanes of electrical connections for a highly distributed high-capacity memory.

In general, the high capacity, fast access time and high endurance available in a QVM circuit, coupled with the high-bandwidth provided by wafer-bonding such a QVM circuit to a processor circuit, enables a powerful electronic device with a high-capacity memory that functions effectively as a large embedded memory, despite the fact that the QVM circuit physically resides on a different wafer or die and not embedded within the semiconductor die on which the processor circuit is formed. The present invention enables or provides significant advantages in many applications, including, for example, artificial intelligence. The electronic devices of the present invention are shown to provide higher bandwidth and lower power than conventional DRAM-based von-Neuman architecture processor systems of comparable memory access times.

FIG. 1a shows a floor plan of semiconductor die 360 that includes memory circuit 355 made up of building blocks referred herein as tiles. In the description herein, each tile can be configured to be individually and independently addressed ("tile-specific basis"). At the user's option, larger memory segments (e.g., a row of tiles or a 2-dimensional block of tiles) may be created and configured to be addressed together (e.g., "row-specific" addressing, or "core-specific" addressing). In any of these organization, the addressable unit may be referred to as a "bank," so that the addressing scheme may be described as "bank-specific". FIG. 1a shows memory circuit 355 being divided into two cores 355a and 355b, each core being a 32-rowx32-column array of tiles in this instance. Cores 355a and 355b sharing local data center 356, where circuits for data selection and for connections to support circuitry may be provided. Examples of support circuitry include error-correction encoders and decoders, address decoders, power supplies, check-bit generators, sense amplifiers, and other circuits used in memory operations. Support circuitry may be formed in the planar semiconductor substrate. In one embodiment, the support circuitry for the thin-film storage transistors of each tile is provided for modularity in the portion of the semiconductor substrate underneath each tile. In FIG. 1a, analog and global driver circuits 357 for signal distribution are formed at one end of semiconductor die 360, and I/O buffer circuits 358 for access to memory circuit 355 is formed at the other end of semiconductor die 360. I/O buffers 358 are provided for sending signals to and receiving signals from an external circuit, when not accessed over the studs. As discussed below, the tiles are modularly designed to include the studs exchanging receiving data and address signals with the wafer-bonded logic circuit over the studs without constraints by I/O buffer 358. FIG. 1a also shows tile 361, which

consists of a 3-D NOR memory array, with the thin-film transistors in each constituent NOR memory string formed along a direction parallel to the planar semiconductor substrate. FIG. 1a shows that the bit lines and global word lines run orthogonally, with local word lines branching-off each global word line and extending along an orthogonal direction perpendicular to the planar semiconductor substrate. As mentioned, the sense amplifiers for each 3-D memory array are formed in the monocrystalline silicon substrate underneath the tile and provide data lines to deliver the output datum.

Although the QVM (and NVM, if present) circuits in the embodiment of FIG. 1a are formed with all the control, sensing, power, input/output and other memory-related functions residing on the memory die itself, it is possible in some other embodiments to have some of these functions physically reside on the processor circuit die. For example, the DDR3/DDR4 or PCIe or other high-speed data interface, or the high voltage transistors required to program or erase the QVM memory, may require analog or high-speed digital transistors and logic circuits that are process-wise incompatible with the thermal budget encountered in the fabrication of the 3-D NOR memory arrays making up the QVM. These circuits may therefore best be placed on the wafer-bonded logic or processor die. The same considerations may apply to other circuits, such as error correcting circuits, arithmetic logic unit circuits, exclusive or circuits, control circuits and state machines. In fact, such circuits may be shared by multiple QVM or NVM dies, and therefore such circuits are most cost-effective at the system level when provided from the processor die over the connections through the stud connectors to each of the individual QVM dies.

FIG. 1b shows system 300, which includes QVM circuit 301 that is wafer-bonded to processor circuit 302 using a flip-chip or flip-wafer technique, according to one embodiment of the present invention. As shown in FIG. 1b, QVM circuit 301 and processor circuit 302 have between them connected studs for two memory buses 303a and 303b, each capable of transferring 2048 bytes (i.e., 2 KB) of data, together with necessary address, check-bits and control signals, per memory cycle. Data transferred over memory buses 303a and 303b, each including close to 20,000 copper connection studs, are processed or prepared in data centers 304a and 304b, respectively. Data center 304a and 304b may also include a memory controller to control memory operations in QVM circuit 301. Computing engine 305, such as a single core or a multi-core processor (e.g., RISC-type processor, such as ARM, or a graphic processor), operates on the data retrieved from or to be written to QVM circuit 301. The high-bandwidth of 4 KB (i.e., 4096 bytes) each memory cycle over memory buses 303a and 303b provides enormous relief to the significant conventional problem of the "Von Neuman bottle neck." With the two memory buses 303a and 303b, simultaneously read and write-back operations can be carried out, which is very beneficial to applications in which a large amount of data are read from memory, processed and written back (e.g., rendering video data). In system 300, processor circuit 302 may also include custom hardware (e.g., AI module 306) for a specific application. For an artificial intelligence application, for example, AI module 306 may include a neural network circuit.

QVM circuit 301 may be configured in numerous ways. For example, FIG. 1c shows memory circuit 320 including 64 rows by 32 columns "core" of tiles, suitable for implementing a portion of QVM circuit 301, in accordance with one embodiment of the present invention. As shown in FIG.

1c, row **63** includes tiles **321-63-0** to **321-63-31** and row **0** includes tiles **321-0-0** to **321-0-31**. In this embodiment, each tile represents an independently addressable 3-D NOR memory array consisting of word lines, bit lines and a number of memory layers stacked one on top of another. In this embodiment, each tile receives and outputs a 536-bit datum, suitable for implementing 512 data bits together with 24 check-bits, or an error-correction encoded 536-bit code word. 536 sense amplifiers are provided in each tile to output the 536-bit output datum on 536 global bit lines that are multiplexed among each column of tiles. The 536 global bit lines (e.g., global bit lines **322-0** to **322-535**) are shared by the 64 tiles in each column, running vertically to connect to data center **323**. In this embodiment, each tile is provided 536 studs to allow parallel access from a wafer-bonded semiconductor die via the bit lines to the thin-film storage transistors of the NOR memory strings of the tile.

Memory circuit **320** thus provides 2048 Bytes of data, along with 768 check-bits, or 2048 Bytes of data in error-encoded code words. As shown in FIG. 1c, adjacent tiles in adjacent rows (e.g., adjacent tiles in rows **62** and **63**) form tile-pairs, with each tile pair consisting of two tiles placed back-to-back (i.e., each being a mirror image to the other). A local bit line is provided for each bit to be stored or to be output from a tile, and a stud is shared between two local bit lines. For example, tile **321-63-0** of row **63** is provided studs **324-63-0** to **324-63-257** and tile **321-62-0** of row **62** is provided studs **324-62-0** to **324-62-257**. In FIG. 1c, data center **323** is formed on the same semiconductor die as the 3-D NOR memory arrays. Alternatively, as shown in FIG. 1b, data centers **304a** and **304b** may provide all or at least some portions of the functions of data center **323**.

It is understood that, although the memory circuit in the embodiment FIG. 1b is described as QVM circuit **301**, such designation is provided merely for illustrative purposes only. The memory of FIG. 1a may have a NVM portion, for example. In one embodiment, selected tiles in the memory circuit are NVM, while other tiles form QVM circuit **301**. In that regard, the wafer-bonding or chip-bonding configurations under the present invention for studs connecting a memory circuit and a processor circuit, including the programmable connectivity disclosed below for QVM, are equally applicable to such memory circuits as DRAMs, phase-change memories (e.g., 3-D XPoint memories), spin-transfer torque (STT) memory devices, resistive random-access memories (RRAMs), or ferroelectric random-access memories (FeRAM).

FIG. 1d is a magnified view of adjacent studs **324-a-n** and **324-(a+1)-(n+1)** for adjacent local bit lines for bits n and $(n+1)$ between two tile-pairs. In one embodiment, each stud may be approximately 1 μm wide or less.

As mentioned above, the studs may be configured to be addressed by all tiles in a row simultaneously or tile-by-tile. FIG. 1e illustrates tile-by-tile studs programmability at the memory tile-level and data path selection at the data center-level, in accordance with one embodiment of the present invention. FIG. 1e illustrates the studs at memory tile row n , having tiles **350-n-0**, **350-n-1**, . . . **350-n-31**, and row $(n+1)$, having tiles **350-(n+1)-0**, **350-(n+1)-1**, . . . **350-(n+1)-31**, respectively. The studs in each tile are driven from or received into the addressed memory cells of each tile at the I/O of line of sense amplifiers at the tile. Thus, the required driver is merely those between on-chip logic gates, which are much smaller than those required for conventional I/O pads, such as those required at the interfaces at each channel of an HBM. Further, in one embodiment of the present invention, the tiles of each row may be configured to be

addressed tile-by-tile, in which case, the 512-bit datum of each tile (536-bit with error-correction coding or check-bits) may be directly driven onto, or receive from, 512-bit (536-bit) data bus **351-a** or **351-b** at the data center (e.g., data center **304a** or **304b**), for example. Alternatively, in one embodiment, selection logic may be configured to allow a data path to be selected for each of data bus **351a** or **351b**. In that configuration, at each row of tiles, one of 32 data paths may be selected to steer one of the 32 536-bit data from its tiles to data bus **351a** or **351b**. The configuration for tile-by-tile addressing or data path addressing may be achieved using, for example, anti-fuses or by a network of transmission gates each controlled by a configuration cell (e.g., a one-bit memory cell) holding a configuration bit. The studs may be made field programmable by allowing user write access to the configuration cells. Alternatively, programming may be achieved during manufacturing using a masking layer.

The flexibility in organizing and programming the stud structure allows multiple data paths to be configured between the memory and logic device, so that multiple memory accesses can take place concurrently, thus providing for overlapped operation. Furthermore, the interconnections and stud routing can be organized in any structure. For example, the input and output signals of any group of tiles are not constrained to be organized as a bus structure of a determined width, be multiplexed for sharing among the tiles, or share any common signaling scheme (e.g., sharing of address and data conventions). There is no restriction on data formats or data types that can be communicated between the logic circuit and the memory circuit, such that there is essentially arbitrary large connectivity with very high bandwidth.

Figure if is a schematic diagram illustrating configuration of studs, according to one embodiment of the present invention. As shown in FIG. 1f, transmission gate network **373** allows interconnection of signals to the studs, allowing user-specified signals to be brought in over the studs and be connected into an array of signals in the memory circuit. Configuration logic **372** to allow configuring portions of the studs for input and for output (e.g., from the I/O line of sense amplifiers) signals, respectively. In addition, configuration cells **370** may be provided to set one of various organizations of the memory tiles (e.g., tile-specific, row-specific, bank-specific, or core-specific addressing, see below). Other organizations and programmable units (e.g., multiple tiles may be logically combined to create a larger addressable memory unit) are possible. The configured memory organization can thus respond the address signals in the desired manner.

The configuration scheme illustrated in Figure if may be provided on both memory circuit **301** and logic circuit **302**, so as to allow any input or output control signal, data signal or address signal to be routed between the two circuits, as desired.

FIG. 1g shows memory circuit **340**, which includes cores **325-1** and **325-2** sharing data center **326**, suitable for implementing a portion of QVM circuit **301**, in accordance with one embodiment of the present invention. In memory circuit **340**, each of cores **325-1** and **325-2** may be a 64 rows $\times$ 32 columns core of tiles, as in the core in memory circuit **320** of FIG. 1c. Data center **326** may be provided similar to data center **323** of FIG. 1c, except that data center **326** is shared between two memory arrays, each of which having 64 $\times$ 32 tiles. In this configuration, an access to a 2K-Byte datum may be delivered at the studs of a row of tiles in 100 nanoseconds or less from each memory array. As cores

325-1 and **325-2** may be accessed simultaneously, 4K-Bytes of data may be delivered every 100 nanoseconds or less. In some embodiments, cores **325-1** and **325-2** of memory circuit **340** form two memory banks.

FIG. 1*h* shows a multi-die configuration implemented using the “flip-chip” technique, in conjunction with a silicon interposer, according to one embodiment of the present invention. As shown in FIG. 1*h*, multiple semiconductor dies **392**, **393**, **394**, **395** and **396** are each wafer-bonded using, for example, the “flip-chip” technique to silicon interposer **391**, which provides a conductor network that interconnects the studs in the semiconductor dies through the studs of silicon interposer **391**. (The interconnection network in silicon interposer **391** are connected through its own studs exposed on its surface with studs in the semiconductor dies). In one embodiment, semiconductor die **396** is a memory circuit, while semiconductor dies **392-395** are each a logic circuit (e.g., each including a RISC-type processor). In this configuration, each logic circuit is provided access to a high-bandwidth, high-capacity memory. In another embodiment, semiconductor die **396** is a logic circuit (e.g., including a RISC-type processor), while semiconductor dies **392-395** are each a memory circuit. In this configuration, the logic circuit is provided access to multiple high-bandwidth, high-capacity memory circuits, forming a computing device with a “mega-embedded memory.” Of course, semiconductor dies **392-396** may be any combination of memory and logic circuits, as optimized for a specific application. For example, one of semiconductor die **392-396** may include a memory controller that manages the configuration or configurations of the memory circuits in the other semiconductor dies.

FIG. 1*i* shows system **380** in which multiple memory circuits **301a** to **301-m** are wafer-bonded to processor circuit **302** using both the flip-chip and the TSV techniques. In this manner, an even higher capacity embedded memory may be made available to the processor or processors in processor circuit **302**. Of course, in the system **380**, only the top memory circuit (i.e., memory circuit **301a**) may be bonded to processor circuit **302** to enjoy the advantages of high capacity, high bandwidth and fast-access to memory circuit **301a**. The other memory circuits **301b** to **301m** are connected through the TSV technique and accessed over a bus structure.

Whether by interconnection studs or by TSVs, when two semiconductor dies are connected, missed connections for any of various reasons are possible. This type of failures is very costly, as wafer-bonding is performed after circuitry on both the bonded semiconductor dies have been completely fabricated. The present invention provides a routing scheme that allows recovery from such a failure. The routing scheme is illustrated in FIG. 7(c). FIG. 7(c) shows rows **721** and **722** of bonding pads on semiconductor dies **723** and **724**, respectively, which are to be wafer-bonded in accordance with the manners described above. In FIG. 7(c), bonding pads in row **721** are labelled **721-1**, **721-2**, . . . , **721-n**. Likewise, bonding pads in row **722** are labelled **722-1**, **722-2**, . . . , **722-n**. In addition, spare bonding pads **721-s** and **722-s** are provided in row **721** and **722**, respectively. Each bonding pad is associated with an interconnection stud or TSV. Bonding pads **721-1** to **721-n** and spare bonding pad **721-s** are each connected to common conductor (“bus”) **727** by a corresponding one of switches **725** (e.g., each a transmission gate, labelled in FIG. 7(c) as transmission gates **725-1**, **725-2**, . . . , **725-n** and **725-s**). Likewise, bonding pads **722-1** to **722-n** and spare bonding pad **722-s** are each connected to common conductor **728** by a corresponding one of switches

726 (e.g., each a transmission gate, labelled in FIG. 7(c) as transmission gates **726-1**, **726-2**, . . . , **726-n** and **726-s**).

If one of the interconnection studs or TSV fails—say, interconnection stud or TSV associated with bonding pad **721-2**, for any reason—corresponding transmission gates **725-2** and **725-s** and transmission gates **726-2** and **726-s** (i.e., their counterparts on semiconductor die **724**) are turned conducting to short bonding pads **725-2** and **726-2** to spare bonding pads **725-s** and **726-s**, respectively. If the interconnection stud or the TSV associated with bonding pads **721-s** and **722-s** are functioning, they provide an alternative signal path to recover from the failure in the interconnection stud or TSV associated with bonding pad **721-2** or **722-2**.

The scheme illustrated in FIG. 7(c) allows recovery for a single interconnecting stud or TSV failure in each row of bonding pads. FIG. 7(d) shows an expanded scheme that provides recovery for any two failures in each row of bonding pads by providing an additional row of transmission gates and an additional common conductor. In FIG. 7(d), row **732** of switches and common conductor **732** provide an alternative routing to recover from a single failure associated with any one of the bonding pads in row **731**, and additional row **733** of switches **733** and common conductor **735** provide an additional alternative routing to recover from an additional failure associated with an additional one of bonding pads in row **731**.

FIG. 2 is a block diagram of memory system **100**, which includes a memory circuit **101** formed on semiconductor substrate that is joined by wafer-bonding to controller circuit **102** formed on a second semiconductor substrate, according to one embodiment of the present invention. As shown in FIG. 2, memory circuit **101** includes memory arrays organized as memory banks **103-0**, **103-1**, . . . , **103-n** and **103-(n+1)**. Control logic circuits **104-0**, **104-1**, . . . , **104-n** and **104-(n+1)** are associated respectively with memory banks **103-0**, **103-1**, . . . , **103-n** and **103-(n+1)** to provide control functions, such as address decoding and timing control for read, write and erase operation sequences. The data read from and to be written into a memory bank resides on internal data buses **106** and **107** respectively. Input/output circuit **105a** steers the data from data bus **106** onto memory bus **110** or steers the data from memory bus **110** to data bus **107**, as required. Memory bus **110** may be provided by numerous connector studs across the wafer-bond between memory circuit **101** and controller circuit **102**. These studs may be formed, for example, by metallic copper. The operations of control logic circuits **104-0**, **104-1**, . . . , **104-n** and **104-(n+1)** and input/output circuit **105a** are controlled by control signals **109**, also driven from state machine **108** in controller circuit **102** over the studs across the wafer bond between memory circuit **101** and controller circuit **102**.

In controller circuit **102**, input/output circuit **105b** operates in a cooperative fashion with input/output circuit **105a** in memory circuit **101** to coordinate signal flows across the studs of memory bus **110**. In this example, memory bus **110** accommodates 64 data bits per bus cycle. FIG. 2 shows that controller circuit **102** includes state machine **108**, data processing circuit (“data center”) **109** and external interface **111**. External interface **111** may be, for example, a memory bus conforming to an industry standard, such as DDR4, DDR5 and PCIe. For purposes of illustration only, data center **109** includes bus **112**—which accommodates two 256-bit pages of data, together with a number of address and command bits—for communication over external interface **111**. For data received from external bus **111** to be written into memory circuit **101**, data center **109** encodes the incoming data into a number of error-correcting code bits

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(e.g., 536 bits from 512 bits of incoming data). In FIG. 2, 64 data bits are communicated over memory bus 110 each bus cycle. Other functions not illustrated in FIG. 2 may be carried out in data center 109. For example, data received from memory circuit 101 may be error-corrected according to the retrieved error correction codes, before being sent to a host device over external bus 111.

The approach of memory system 100 of FIG. 2 can be extended to other applications, such as a storage system controller. FIG. 3a shows storage system 200 in the prior art for a storage system that includes storage device 204 with flash cache memory 203. As shown in FIG. 3a, storage system 200 includes storage control circuit 201, which controls the operations of storage device 204 with flash cache memory 203. Typically, flash cache memory 203 includes numerous flash memory arrays of NAND memory strings which operations are controlled over a NAND interface by storage controller 207, which also controls storage device 204. Typically, storage control circuit 201 also includes (i) storage interface 205, which interfaces with a host device over storage interface 211 (FIG. 3a), and (ii) memory controller 206, which provides access to a fast run-time memory system, such as dynamic random-access memory (DRAM) 202.

Typically, when data is received over storage interface 211 from the host device ("write data"), or when data is read from either flash cache memory 203 or storage device 204 by storage controller 207 ("read data"), the data is placed in DRAM 202. Central processing unit (CPU) 208's role in managing read and write accesses to storage device 204 and flash cache memory 203 and in maintaining cache concurrency are well-known to those of ordinary skill in the art. In addition, CPU 208 also manages false erase, wear leveling and translation table locator operations, as also known by those of ordinary skill in the art.

The high capacity and fast access times of the QVM and NVM of the Non-provisional Applications, as well as the high endurance of the QVM, together with the high bandwidth achievable using wafer-bonding, enable the new applications of the present invention. FIG. 3b shows storage system 200 that includes NVM and QVM circuits 210 that are wafer-bonded with multi-interface controller circuit 209, according to one embodiment of the present invention.

In FIG. 3b, NVM and QVM circuits 210 may be formed on the same semiconductor die or on different semiconductor dies. Wafer-bonding of NVM and QVM circuits 210 to multi-interface controller circuit 209 may be achieved preferably using the flip-chip technique. However, any multi-die configuration using any of TSV, silicon interposer or silicon bridge techniques may also be possible. In storage system 200, NVM and QVM circuits 210 and multi-interface controller circuit 209 replace DRAM 202 of FIG. 3a, as the QVM circuit can offer a higher capacity at a lower cost than DRAM and requires less power. In storage system 200, relative to storage controller 207, multi-interface controller circuit 209 can serve as a NAND controller to manage flash cache memory 203 more effectively, as translation tables, a directory, file allocation tables (FATs) and journaling files may be stored in the NVM circuit or the QVM circuit. Multi-interface controller circuit 209 can access highly efficiently and simultaneously to both NVM and QVM circuits 210 and flash cache memory 203. The higher capacity allows larger translation tables, while fast access to flash cache memory 203 allows more efficient flash management operations (e.g., "erase ahead"), which are not possible in storage system 200 of the prior art. Furthermore, multi-interface controller circuit 209 can monitor flash cache

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memory 203 for "hot files," which can be moved into the NVM circuit to allow faster accesses and reduce accesses to flash cache memory 203, thereby improving endurance performance. Other functions, such as generating error-correction codes for all the NVM and QVM circuits, as well as for hard-disk system 204, may also be carried out in multi-interface controller circuit 209.

FIG. 3c shows one implementation 270 of NVM and QVM circuits 210 and multi-interface controller circuit 209, according one embodiment of the present invention. As shown in FIG. 3c, multi-interface controller 209 includes NAND interface 251 for controlling flash cache memory 203, DRAM interface 261 (e.g., a data bus conforming to DDR3 or DDR4 standard), and NVM circuit 210a and QVM circuit 210b, corresponding to the NVM and QVM portions of NVM and QVM circuits 210, respectively. In implementation 270, as NVM and QVM circuits 210 are wafer-bonded to multi-interface controller circuit 209, interfaces to NVM circuit 210a and QVM circuit 210b are implemented by the studs provided in the semiconductor dies. One or more reduced instruction set (RISC)-type processors provide operational control of multi-interface controller circuit 209. As shown in FIG. 3c, implementation 270 includes DMA controller 253 for coordinating reading and writing data among NVM circuit 210, QVM circuit 210b and flash cache memory 203. Address and data to be sent or received over DRAM interface 261 may be buffered at address/data buffering circuit 257. In addition, check-bit generation circuit 256 and error-correction code circuit 255 are provided to perform the conventional data integrity functions. Timing signals may be provided from reference generator 254.

RISC-type processor 252 manages the storage controller functions, such as translation tables and cache tables. In addition, multi-interface controller circuit 209 may also participate in managing storage device 204 (see, FIG. 3b). For example, upon receiving a write command over storage interface controller 205 (FIG. 3b) for a write operation to storage system 203, a buffer in QVM circuit 210b is allocated. The write data is loaded into a buffer in QVM circuit 210b, together with check-bits generated from check-bit generation circuit 256. If industry standard FAT format is used, the data is sent to storage system 204 in blocks at designated locations. Flash cache memory 203 and NVM circuit 210a are checked for a cached copy of the file. Such a cached copy is then updated by initiating a parallel write, so as to maintain cache concurrency. An access frequency is maintained for each file to enable wear-leveling and other purposes. A suitable file relocation policy may be implemented.

Similarly, when a read command to storage device 204 is received over storage interface 211, buffer space is allocated in QVM circuit 210b. If a copy of the requested file is present in flash cache memory 203, no access to storage device 204 is initiated. A cache translate table in NVM circuit 210a identifies the location of the file in flash cache memory 203. The requested file is then read into QVM circuit 210b, after error-correction and generation of check-bits, if necessary. If there is not a cached copy, the FAT table entry in NVM circuit 210a for the file is parsed to create a scatter gather-list in one part of the buffer space. Storage controller 207 fetches the data from storage device 204 and provides the read data over DRAM interface 251. Address/data buffering circuit 257 places the data in the buffer space allocated in QVM circuit 210b, after error-correction and together with generated check-bits. Once the data is in QVM circuit 210b, DMA controller 253 coordinate data transfer over storage interface 211 (FIG. 3b).

As mentioned above, QVM circuit **210a** may have a data retention time that requires a refresh operation (e.g., every 10 minutes). In one embodiment, QVM circuit **210b** is divided into refresh zones. When data is transferred into a refresh zone, an “active bit” is marked in a refresh table for the refresh zone to be refreshed on a periodic basis. When data is transferred out of a refresh zone, its corresponding “active bit” is reset, which halts refresh operation in that refresh zone. This refresh scheme is particularly advantageous in many applications, such as those that use the QVM circuit as temporary or transitory storage for read or write operations from or to a disk system or an NVM storage device and for receiving or passing data to a server DRAM. When the temporary or transitory operation is completed, no refresh to allocated space in that portion of memory is necessary. Furthermore, in very active usage (e.g., when files are moved quickly), such that no data of the transitory operation stay in the QVM memory for a greater period than the refresh interval, no refresh is required. In data flow applications, only a very small number of refreshes is likely triggered, which results in low power operation and higher performance.

Other embodiments of the present invention are applicable to field-programmable logic devices, such as field-programmable gate arrays (FPGAs). In this description, while FPGAs are used as examples for illustrative purposes, it is understood that the same principles are applicable to virtually all field-programmable logic devices. FPGAs are typically fabricated under a logic process, which is usually incompatible for forming on the same semiconductor substrate large capacity or high-performance memory circuits. Therefore, FPGA designs typically include neither non-volatile memory nor DRAM memory circuitry. Memory resources on an FPGA circuit are typically provided by fast static random-access memory (SRAM). As SRAM circuits are volatile, an FPGA circuit must be configured upon power-up from an external circuit, typically from a non-volatile memory circuit. The bandwidth for transferring data for configuration of an FPGA circuit, often provided over a serial interface, is the bottle-neck that creates a lengthy latency to the FPGA circuit becoming functional during a power-on reset.

FPGA circuits often include other application-specific custom logic circuits on the same semiconductor die, such as industry standard data interfaces (e.g., PCIe, DDR3/4 and other interfaces), ethernet cells, processor cores and neural networks.

One long-felt need for designers using FPGA circuits is the availability of a high-capacity, high bandwidth memory. The present invention addresses this long-felt need by wafer-bonding a high-capacity memory circuit (e.g., those described in the Non-provisional applications) with an FPGA circuit, taking advantage the high-bandwidth, low-power data transfer over the studs. FIG. **4a** shows system **400**, in which high-capacity memory circuit **401** is wafer-bonded using the flip-chip technique to FPGA circuit **402**, according to one embodiment of the present invention.

In memory circuit **401**, portion **401a** implements NVM circuits, while portion **401b** implements QVM circuits. While FIG. **4a** shows memory circuit **401** as including both QVM and NVM circuitry, other embodiments may have QVM and NVM circuitry provided separately on separate semiconductor dies, wafer-bonded using, for example, any TSV, silicon interposer or silicon bridge techniques, or a combination of these techniques. In system **400**, data from NVM portion **401a** and QVM portion **401b** are communicated over data buses **403** and **404**—implemented by wafer-

bonded studs—to FPGA circuit **402**. The data transfers over data buses **403** and **404** are controlled by NVM data center **408** and QVM data center **409** that are implemented on FPGA circuit **402**. FPGA circuit **402** includes programmable logic circuitry **407a** and corresponding configuration circuitry **407b**, and custom logics **405** and **406**. For example, custom logic circuit **405** may be an industry standard data interface (e.g., PCIe), while custom logic circuit **406** may be one or more RISC-type or graphics-type processor cores, or arithmetic-logic units (ALUs).

Providing data centers **408** and **409** on FPGA circuit **402** allows a flexibility that is advantageous in many configurations and applications. For example, FPGA circuit **402** may be configured to implement numerous functions that may have memory requirements or have usage patterns that are independent of each other. This application may be satisfied, for example, by configuring the memory in memory circuit **401** into dual-ported segments that can allow read and write-back operations in each segment independently of other segments. Allowing the memory circuit **401** to be configured in flexible ways from FPGA circuit **402** enable efficiency or make possible many applications. For example, segmented processing provides advantages for applications such as neural networks. Also, rather than moving large amount of data (“data sets”) required for calculations, the data sets may be stored in different data segments and swapped for processing to take advantage of the quick context switching. Furthermore, the memory segments can be configured into one or more first-in-first out (FIFO) segments suitable for pipelined applications.

(Note that, although FIG. **4a** uses as an example an FPGA circuit, the principles described herein for FPGA circuit **402** is applicable to other applications, such as a dedicated controller connected to memory circuit **401** in this manner).

FIG. **4b** is a logical representation of a configuration in which QVM portion **401b** of memory circuit **401** is segmented into memory segments **421-1**, **421-2**, . . . , and **421-n**, respectively serving corresponding segments of programmable logic (“tile cells”) **422-1**, **422-2**, . . . , and **422-n** over data buses **423-1**, **423-2**, . . . , and **423-n**. In FIG. **4b**, data buses **423-1**, **423-2**, . . . , and **423-n** are each provided by the studs in the wafer-bond between memory circuit **401** and FPGA circuit **402**. QVM portion **401b** may be of modular design, such that adjacent segments may be combined to form larger segments for applications requiring additional memory resources. The segmented approach obviates the need for data transfer using complex circuitry (e.g., DMA or data connect center) that requires allocation of buffer memory and other circuitry that may incur substantial delay. QVM (or NVM) memory **421-1** to **421-n** may be segmented, according to one embodiment of the presentation, in part by providing dedicated control and sensing logic circuitry associated with each memory segment, thus making each memory segment semi-autonomous. Such control and sensing circuitry is most efficient when implemented either below or above the memory layers of the segment. Under the segmented approach, the bandwidth available to studs allocated to a memory segment is dedicated to the segment alone and no buffering is required. The segmented design provides great advantage in applications, such as neural networks, in which large number processors operating simultaneously have need for frequently fetching and writing back data. Segmentation may be configured using a configuration scheme similar to that shown in FIG. **1f**, for example.

The segmented approach which allows multiple parallel electrical connections between the logic device and the memory device, provides for many memory usage schemes

not previously possible. For example, by partitioning the memory into smaller blocks that are logically concatenated, a user may use the memory as a dual ported device. In that usage scheme, when writing an amount of data spanning two or more small blocks, after the first small block is written and while the system is writing the remainder of the data to the next small block, the first block is available for reading. The segmented approach allows a reading operation to initiate before all the data is written. Allowing initiation of reading to overlap with the write operation on the rest of the data block being transferred saves time. In some DMA operations, this ability results in very high burst rates. In the prior art, in large transfers (e.g., 10 GB), some systems use SRAM for bursting, or wait for the transfer to complete before starting high speed transfer out of the space. A burst capability can save cost and support high speed packet transfers before one side of a transfer is complete.

FIG. 4c a logical representation of a variation of the configuration of FIG. 4b, in which tile cell segments 422-1, 422-2, . . . , and 422-*n* are also paired with custom logic circuits. In FIG. 4c, custom logic circuits 424-2, 423-3, . . . , and 424-(*n*-1) are seen operating with tile cell segments 422-2, . . . , and 422-(*n*-1). Such custom logic circuits may include, for example, a RISC-type processor a graphics processing unit. In some embodiment, one of the memory segments may be used as a code store for the custom logic circuit processor. The approach of partitioning memory circuit 401 into memory segments can optimize many applications. For example, FIG. 4d illustrates a neural network application implemented by the configuration of FIG. 4c, in which each memory segment and its associated programmable logic tile cell segment, together with its custom logic, implement a stage in a neural network, in accordance with one embodiment of the present invention.

Under the present invention, both processor-based device (e.g., the electronic device of FIG. 3a) and the FPGA-based device (e.g., the electronic device of FIG. 4a) have great advantage for implementing applications that are both computation and memory centric. The segmented approach allows parallel operations using multiple processors (e.g., RISC-cores) with high-bandwidth access to a high capacity memory. Alternatively, the memory segments can be configured as stages of a pipeline for simultaneous processing by multiple processors each optimized for a specific stage, as illustrated in FIG. 4d. Because of the high bandwidth to the memory accessible by the processing units, very high throughput can be expected as interim results for each stage of the computation are made available to the next stage without delay resulting from the memory bandwidth limitation.

The flip-chip wafer-bonded combinations of memory and FPGA circuits, as provided by the present invention, has the advantage a fast power-on, as the FPGA circuit can be quickly configured at power-on over the high-bandwidth interface from NVM portion 401a of memory chip 401. The high-bandwidth interface allows configuration bits to be set in parallel for a fast configuration to allow the configured programmable logic to be ready shortly after power-on. In some embodiments, a part of the configuration logic may be subsequently configured for functional operation. Having an NVM portion in memory 401 also enables dynamic configuration at run-time, which is a difficult, if not impossible, operation in the prior art.

A field-programming logic device (e.g., an FPGA) may be implemented as a look-up table in a memory circuit. In that implementation, the data input is used an address to the memory circuit and the logic function corresponding to the

data input is stored at the addressed location. Thus, the programmable logic (e.g., the tile cells of FIG. 4c), for example, may be implemented memory circuit (e.g., memory circuit 401 of FIG. 4a), with its output provided over the studs between the memory circuit and a wafer-bonded logic circuit for processing. FIG. 5a shows programmable logic tile cell 500 implemented in 8K×4-byte memory array 501, which can provide an 8-bit output datum directly to an embedded 8-bit processor or a similar logic circuit. In this example, the 8-bit output datum is selected in multiplexer 502 by the address bits and output by 3-state driver 503 (or direct drive).

With a suitable modular design, multiple programmable logic tile cells can be configured to form a larger programmable logic circuit. For example, FIG. 5b shows programmable logic tile cells 501a and 501b in a memory circuit configured to provide to make a 16K×4-byte memory array.

To implement programmable logic in memory, configurable logic must be provided to allow configuration of the input signals, the output signals and the address signals. Such configurable logic may be implemented by transmission gates programmed by configuration cells.

In some instances, it may be advantageous to allow arithmetic and logic operations in a modular memory segment, which may be useful in many applications. ALUs are known to those of ordinary skill in the art. Alternatively, arithmetic operations can often be carried out using a multiply-accumulate unit (MAC). MACs are also known to those of ordinary skill in the art. FIG. 5c shows modular memory segment 570 that has associated ALU 575 for arithmetic and logic operations, in accordance with one embodiment of the present invention. As shown in FIG. 5c, memory segment 570 includes memory tiles 540-0 to 540-31, each containing 536 error-correction coded bits, similar to those described above, such as memory tile 324-0-0 of FIG. 1c. The 536-bit output of each tile is processed at a corresponding one of error correction modules 541-0 to 541-31 to provide a 512-bit error-corrected datum, which may be latched into a corresponding one of 512-bit register 542-0 to 542-31. Any of the data bits in 512-bit registers 542-0 to 542-31 may be selected by a corresponding one of 512:1-multiplexers 543-0 to 543-31 into a corresponding bit in one of the two input registers A and B of 32-bit ALU 545. Error-correction modules 541-0 to 541-31, 512-bit registers 542-0 to 542-31, 512:1-multiplexers 543-0 to 543-31, registers A and B and ALU 545 may all be placed in one of the data centers of the memory circuit, for example.

The QVM circuits of the Non-provisional Applications may be formed out of various deposited materials over a silicon substrate. In some embodiments, the 3-dimensional memory arrays of the QVM circuits, even with multiple layers of memory cells (e.g., 8, 16 or 32), are generally less than a few microns thick. According to one embodiment of the present invention, conductors may be provided in vias that run the entire thickness of the memory array ("through vias"). With its silicon substrate removed, the QVM circuit may be wafer-bonded on both of the opposite sides to two other circuits. In such a configuration, the conductors in the through vias may be used to connect these wafer-bonded other circuits. Unlike conventional TSVs—which are typically hundreds of microns thick—the through vias of the present invention are merely a few microns thick, thus they are easier to fabricate relative to TSVs and with a much lower resistance. FIGS. 6a-6d illustrate the use of through vias in a QVM circuit that is wafer-bonded with two other circuits in accordance with one embodiment of the present invention.

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FIG. 6a shows a cross section of QVM circuit 601 and logic circuit 602 (e.g., a memory controller circuit). Logic circuit 602 may include, for example, registers, buffer circuits, and memory operation support circuitry 639 for programming, erasing and reading the memory arrays of QVM 601 formed in or on silicon substrate 638. In addition, memory operation support circuitry 639 may also include power supply circuitry. QVM circuit 601 may include QVM memory array structure 617 formed above silicon substrate 618. In FIG. 6a, hybrid bonding pads 624 and 631 are provided at the surface of logic circuit 602 for electrical access to memory operation support circuitry 639 formed at or on silicon substrate 638. Memory array structure 617 is generally embedded in silicon oxide. Although shown in FIG. 6a, merely for illustrative purpose only, as having only two layers of memory cells, QVM memory array structure 617 may include any number of memory cell layers.

In FIG. 6a, memory array structure 617 includes memory cells portion 616 and “staircase” portion 620. Memory cells portion 616 may include, for example, NOR memory strings. The common bit line and the common source line of each NOR memory string may be electrically accessed from staircase portion 616. (In some embodiments, as disclosed in the Non-provisional Applications, the common source lines need not be externally accessed.) FIG. 6a shows the bit and source lines are connected, in staircase portion 620, by vias to interconnect layers 614, which are connected by vias to “hybrid” bonding pads 611 at the top surface of QVM circuit 601. Interconnect layers 614 are sometimes provided as “redistribution layers,” as known to those of ordinary skill in the art. Local word lines to the NOR memory strings are provided vertically in memory cells portion 616 and routed through two or more layers of interconnect conductors (“global word lines”) provided below and above memory array structure 617. The global word lines are also connected by vias to hybrid bonding pads 611 at the surface of QVM circuit 601. Hybrid bonding pads 611 are also provided for routing power and ground supply voltage lines. FIG. 6a also shows QVM 601 includes conductor-filled through vias 613 that substantially run the entire thickness of memory array structure 617. Hybrid bonding pads 610 are provided at the top surface of QVM circuit 601 to access conductor-filled through vias 613.

FIG. 6b shows QVM circuit 601 “flipped over” and wafer-bonded with logic circuit 602. The wafer-bonding may be achieved using, for example, “hybrid bonding” techniques that effectuate bonding of the corresponding silicon oxide materials and the corresponding copper materials across the surfaces of QVM circuit 601 and logic circuit 602. As shown in FIG. 6b, hybrid bonding pads 610 (connected to through vias 613) on QVM circuit 601 are bonded to hybrid bonding pads 624 of logic circuit 602 and hybrid bonding pads 611 of QVM circuit 601 are bonded to hybrid bonding pads 631 of logic circuit 602.

Thereafter, silicon substrate 618 of QVM circuit 601 is substantially removed, for example, by grinding. In FIG. 6c, substantial removal of silicon substrate 618 exposes the silicon oxide in which conductor-filled through vias 613 are embedded. Vias 640 are then etched into the silicon oxide of QVM circuit 601 and conductor-filled. (Suitable conductor material may be, for example, copper). Then, as shown in FIG. 6d, redistribution layer 651 including copper interconnection network 652 may be provided by a damascene process, in advance of attaching a second wafer-bonded circuit to QVM circuit 601. Interconnection network 652 is made available for access electrically by bonding pads 653. In FIG. 6d, the second wafer-bonded circuit is shown as

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application-specific integrated circuit (ASIC) die 660. ASIC die 660 may be bonded to QVM circuit 601 using die-to-wafer bump bonds, as known to those of ordinary skill in the art.

The above detailed description is provided to illustrate specific embodiments of the present invention and is not intended to be limiting. Numerous variations and modifications within the scope of the present invention are possible. The present invention is set forth in the accompanying claims.

We claim:

1. An integrated circuit assembly, comprising:

a first integrated circuit die having (i) a semiconductor substrate having circuitry formed therein or formed at a planar surface of the semiconductor substrate; (ii) an interconnection layer formed above the planar surface, the interconnection layer having conductors that are configurable to electrically connect to the circuitry; (iii) an insulation layer formed over the planar surface semiconductor substrate and encapsulating the interconnecting layer; and (iv) one or more bonding pads exposed on a surface of the insulation layer, wherein the insulation layer has formed therein conductor-filled vias that electrically connect between the conductors in the interconnection layer and the bonding pads; and

a second integrated circuit die having an insulation layer of a predetermined thickness, the insulation layer encapsulating (i) a plurality of modular memory circuits, each comprising one or more 3-dimensional arrays of storage transistors; (ii) an interconnection layer of conductors; (iii) bonding pads exposed on a first surface and on a second surface of the insulation layer, wherein a portion of the bonding pads on the first surface are bonded to bonding pads of the first integrated circuit die; (iv) conductor-filled through vias each electrically connecting a bonding pad on the first surface to a corresponding bonding pad on the second surface; and (v) a data center circuit that configure the conductors of the interconnection layer to route data signals and control signals from one or more groups of the modular memory circuits to the conductor-filled through vias.

2. The integrated circuit assembly of claim 1, wherein the second integrated circuit die further comprises a semiconductor substrate on which the modular memory circuits are formed.

3. The integrated circuit assembly of claim 2, further comprising a third integrated circuit die having (i) a memory or logic circuit formed therein or thereon, (ii) an interconnection layer of conductors formed above the memory or logic circuit; (iii) an insulation layer encapsulating the interconnection layer and the memory or logic circuit; and (iv) bonding pads exposed on one surface of the insulation layer, a portion of the bonding pads being bonded to those bonding pads on the second surface of the second integrated circuit die that are electrically connected to one of the conductor-filled vias, wherein the interconnection circuit is configurable to selectively electrically connect data signals and control signals of the memory or logic circuit to the bonding pads of the third integrated circuit die.

4. The integrated circuit assembly of claim 3, wherein the third integrated circuit die is attached to the second integrated circuit die by a die-to-wafer bump bonding technique.

5. The integrated circuit assembly of claim 3, wherein the third integrated circuit die is attached to the second integrated circuit die by a wafer bonding technique.

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6. The integrated circuit assembly of claim 1, wherein the circuitry formed on the semiconductor substrate of the first integrated circuit die comprises support circuitry that operates the modular memory circuits in the second integrated circuit die.

7. The integrated circuit assembly of claim 6, wherein the support circuitry comprises sense amplifiers, voltage sources for generating operating voltages, and a state machine for managing read, write, erase, and refresh operations at selected modular memory circuits.

8. The integrated circuit assembly of claim 6, wherein under control of the support circuitry, two or more of the modular memory circuits are operated in parallel.

9. The integrated circuit assembly of claim 6, wherein the support circuitry comprises two or more memory controller circuits.

10. The integrated circuit assembly of claim 9, wherein each memory controller circuit comprises a programmable microprocessor.

11. The integrated circuit assembly of claim 9, wherein the first integrated circuit die further comprises a host interface accessible by each memory controller for communicating with a host device, and wherein each memory controller further comprises logic circuits configured to implement management functions of modular memory circuits, one or more write buffers for storing write data to be stored in the modular memory circuits, and an error correction circuit for performing error correction on data stored in the modular memory circuits.

12. The integrated circuit assembly of claim 11, wherein the host interface conforms to an industry standard interface, being one of: DDR3/DDR4 or PCIe.

13. The integrated circuit assembly of claim 11, wherein each memory controller circuit further comprises one or more data processing circuits each processing data to be stored into or read from a first corresponding group of modular memory circuits.

14. The integrated circuit assembly of claim 13, wherein each data processing circuit further processes data for a second corresponding group of modular memory circuits, and wherein the first corresponding group of modular memory circuits is placed adjacent the second corresponding group of modular memory circuits in the second integrated circuit die.

15. The integrated circuit assembly of claim 13, wherein each data processing circuit comprises one or more of: error-correcting circuits, check-bit generation circuits, registers, arithmetic logic units, multiplexers and multiply-accumulate circuits.

16. The integrated circuit assembly of claim 1, wherein each modular memory circuit comprises a word line driver circuit and a bit line driver circuit.

17. The integrated circuit assembly of claim 1, wherein the data center circuit comprises address decoders for selecting designated modular memory circuits.

18. The integrated circuit assembly of claim 1, wherein the storage transistors in the modular memory circuits each have a retention time of greater than 1 day and an endurance of a million or more program-erase cycles.

19. The integrated circuit assembly of claim 1, wherein modular memory circuits each comprises a non-volatile memory circuit.

20. The integrated circuit assembly of claim 1, wherein the storage transistors in the modular memory circuits each comprises a ferroelectric storage transistor.

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21. The integrated circuit assembly of claim 1, wherein the 3-dimensional arrays of storage transistors are each organized as a plurality of NOR memory strings.

22. The integrated circuit assembly of claim 1, wherein the first and the second integrated circuit dies are wafer-bonded using a flip-chip technique.

23. The integrated circuit assembly of claim 1, wherein the modular memory circuits are arranged along a plurality of rows and a plurality of columns.

24. The integrated circuit assembly of claim 23, wherein the modular memory circuits are configured according to a memory segmentation scheme into memory segments that are independently addressable (a) by modular memory circuits individually, (b) row-by-row, or (c) block-by-block, wherein each block of memory units consists of modular memory circuits within a predetermined number of rows and a predetermined number of columns.

25. The integrated circuit assembly of claim 24, wherein the data center circuit comprises a plurality of configuration cells, a plurality of signal paths that connect control, address and data signals of the modular memory circuits to their respective groups of conductors according to the memory segmentation scheme, the signal paths being configured by values stored in the configuration cells.

26. The integrated circuit assembly of claim 25, further comprising a network of switches each controlled by one of the configuration cells.

27. The integrated circuit assembly of claim 25, wherein the configuration cells are field-programmable.

28. The integrated circuit assembly of claim 25, wherein the signal paths are configured by anti-fuses.

29. The integrated circuit assembly of claim 1, wherein the first and the second semiconductor dies are wafer-bonded using any one of: through silicon-via, silicon interposer, silicon bridge techniques and combinations thereof.

30. The integrated circuit assembly of claim 1, wherein each modular memory circuit further comprises (i) programmable logic circuits in the form of look-up tables, and (ii) memory cells storing configuration data, and wherein the look-up tables are configured using the configuration data.

31. The integrated circuit assembly of claim 30, wherein the programmable logic circuits each comprises logic circuits in a configurable neural network.

32. The integrated circuit assembly of claim 31, wherein each programmable logic circuit further comprises processor circuits in the configurable neural network.

33. An electronic device, comprising:

a first semiconductor die comprising a substrate layer with a planar surface, a memory circuit and first support circuitry for operating the memory circuit, wherein:

(a) the memory circuit, which is formed above the planar surface, comprises a plurality of modular memory units, each modular memory unit comprising (i) a three-dimensional array of storage transistors organized into a plurality of memory strings; (ii) a first insulation layer formed over the substrate layer and encapsulating the memory circuit; and (iii) a group of conductors exposed to a surface of the first insulation layer, the group of conductors being configured for communicating control, address and data signals associated the memory unit; and

(b) the first support circuitry is formed at or on the planar surface of the substrate layer underneath the memory circuit; and

a second semiconductor die having formed thereon a second insulation layer encapsulating second support circuitry for operating the memory circuit, the second

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support circuitry including a logic circuit having a plurality of conductors each exposed at a surface of the second insulation layer, wherein the first and second semiconductor dies are wafer-bonded, such that the conductors exposed at the surface of the first insulation layer are each electrically connected to a corresponding one of the conductors exposed at the second insulation layer.

34. The electronic device of claim 33, wherein the first and the second semiconductor dies are wafer-bonded using a flip-chip technique.

35. The electronic device of claim 33, wherein each storage transistor has an endurance of a million or more write-erase cycles.

36. The electronic device of claim 33, the memory circuit further comprises a quasi-volatile memory circuit.

37. The electronic device of claim 36, the memory circuit further comprises a non-volatile memory circuit formed alongside the quasi-volatile memory circuit above the substrate layer.

38. The electronic device of claim 36, wherein the quasi-volatile memory circuit includes memory cells each storing more than one-bit of data.

39. The electronic device of claim 33, wherein the modular memory units are arrayed in a regular configuration.

40. The electronic device of claim 39, wherein the regular configuration comprises a plurality of blocks of modular memory units, each block being organized as a plurality of rows and a plurality of columns, wherein the modular memory units are configured according to a memory segmentation scheme into memory segments that are independently addressable (i) by modular memory unit individually, (ii) row-by-row, or (iii) block-by-block.

41. The electronic device of claim 40, further comprising a plurality of configuration cells, wherein the memory circuit comprises a plurality of signal paths that connect the control, address and data signals of the modular memory units to their respective groups of conductors according to the memory segmentation scheme, the signal paths being configured by values stored in the configuration cells.

42. The electronic device of claim 41, wherein the configuration cells are field-programmable.

43. The electronic device of claim 33, wherein the first support circuitry includes a plurality of circuit groups, wherein each circuit group supports memory operations of an associated one of the modular memory units that is formed above the associated circuit group along a direction substantially normal to the planar surface.

44. The electronic device of claim 43, wherein each circuit group in the first support circuitry comprises address decoders, voltage sources and sense amplifiers.

45. The electronic device of claim 33, wherein the logic circuit comprises voltage sources for generating control signals for programming or erasing the storage transistors of the modular memory units, wherein signals of the voltage sources are provided to each modular memory unit over its associated group of conductors.

46. The electronic device of claim 33, wherein the logic circuit comprises an industry standard host interface over which the memory circuit communicates with a host device.

47. The electronic device of claim 46, wherein the industry standard interface comprises one of: DDR3/DDR4 or PCIe.

48. The electronic device of claim 33, wherein the logic circuit comprises one or more data processing circuits that control operations in the memory circuit, or that processes data read from or to be written into the memory circuit, the

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data processing circuits being one or more of: error correcting circuits, arithmetic logic unit circuits, exclusive or circuits, control circuits and state machines.

49. The electronic device of claim 48, further comprising an additional memory circuit formed on a third semiconductor die also wafer-bonded to the second semiconductor die, and wherein the data processing circuits are shared among the memory circuit and the additional memory circuit.

50. The electronic device of claim 33, wherein the memory circuit comprises one of: DRAMs, phase-change memories, spin-transfer torque (STT) memory devices, resistive random-access memories (RRAMs), or ferroelectric random-access memories (FeRAM).

51. The electronic device of claim 33, wherein the first and the second semiconductor dies are wafer-bonded using any one of: through silicon-via, silicon interposer, silicon bridge techniques and combinations thereof.

52. The electronic device of claim 33, the memory circuit further comprising a data processing circuit that processes data to be stored into or read from one or more of the modular memory units.

53. The electronic device of claim 52, wherein a first group of modular memory units and a second group of modular memory units are placed on opposite sides of the data processing circuit.

54. The electronic device of claim 53, wherein the data processing circuit includes a first portion and second portion placed separated from each other, and wherein a first group of the modular memory units operate with the first portion of the data processing circuit and wherein a second group of the modular memory units operate with the second portion of the data processing circuit.

55. The electronic device of claim 52, wherein the data processing circuit comprises one or more of: error-correcting circuits, check-bit generation circuits, registers, arithmetic logic units, multiplexers and multiply-accumulate circuits.

56. The electronic device of claim 33, wherein the logic circuit comprises a data processing circuit that includes a configuration logic circuit that configures the modular memory units into a selected one of a plurality of memory configurations and wherein the data processing circuit controls data to be stored into or read into the modular memory units according to the selected memory configuration.

57. The electronic device of claim 56, wherein the selected memory configurations include memory segments of various sizes each formed by combining two or more of the modular memory units.

58. The electronic device of claim 33, wherein the logic circuit comprises programmable logic circuits that implement a configurable neural network.

59. The electronic device of claim 58, wherein the logic circuit further comprises processor circuits that implement one part of the configurable neural network.

60. The electronic device of claim 33, wherein the logic circuit comprises a plurality of modular field-programmable logic circuits.

61. The electronic device of claim 60, wherein the modular memory units are configured into first and second groups of modular memory units, wherein the modular field-programmable logic circuits are configured into first and second groups of modular field-programmable logic circuits configured to access data in the first and second groups of modular memory units, respectively.

62. The electronic device of claim 61, wherein each of the first and second groups of modular field-programmable logic circuits implements a neural network.

63. The electronic device of claim 61, wherein each of the first and second groups of field-programmable logic circuits provides output data for processing by one or more custom logic circuits. 5

64. The electronic device of claim 63, wherein each custom logic circuit comprises a RISC-type processor or a graphics processor unit. 10

65. The electronic device of claim 33, wherein the logic circuit comprises a memory controller.

66. The electronic device of claim 33, wherein the conductors exposed at the surface of each semiconductor die are organized into groups and wherein the conductors of each group are each connected by an associated one of a first plurality of switches to a first common conductor. 15

67. The electronic device of claim 66, wherein the conductors of each group are each connected by an associated one of a second plurality of switches to a second common conductor. 20

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